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Freitas Palricas**

**Explorando os Efeitos de Jogos Sérios de
Realidade Virtual Personalizada para
Reabilitação de Acidente Vascular Cerebral**

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Reality Serious Game for Stroke Rehabilitation**



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Dissertação apresentada à Universidade de Aveiro para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Desenvolvimento de Jogos Digitais, realizado sob a orientação científica do Doutor Bernardo Marques, Professor auxiliar do Departamento de Eletrónica, Telecomunicações e Informática da Universidade de Aveiro, da Doutora Beatriz Sousa Santos, Professora catedrática do Departamento de Eletrónica, Telecomunicações e Informática da Universidade de Aveiro.

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palavras-chave

Realidade Virtual, Jogos Sérios, Personalização, Design Centrado no Utilizador, Acidente Vascular Cerebral, Reabilitação

resumo

O Acidente Vascular Cerebral (AVC) é uma das principais causas de incapacidade de longa duração a nível mundial, comprometendo progressivamente as funções motoras, cognitivas e a qualidade de vida dos sobreviventes. Os programas de reabilitação convencionais apresentam duas limitações críticas: a monotonia dos exercícios terapêuticos repetitivos e a ausência de adaptação individualizada fatores que corrompem a motivação e o envolvimento do paciente. Esta dissertação apresenta o Country Fair VR, um protótipo de jogo sério em Realidade Virtual para o Meta Quest 3, concebido para apoiar a reabilitação dos membros superiores e das funções cognitivas de sobreviventes de AVC. O sistema assenta num framework de personalização em *closed-loop* com um motor de Ajuste Dinâmico de Dificuldade baseado em regras, implementando um modelo híbrido que combina adaptação automática em tempo real com a capacidade de sobreposição manual pelo terapeuta, preferência validada por 70% dos profissionais de saúde inquiridos no Centro de Reabilitação de Rovisco Pais. O protótipo inclui dois mini-jogos terapeuticamente fundamentados numa feria popular virtual: um jogo de Tiro com Arco, focado na força bimanual, treino propriocetivo e atenção seletiva, e um jogo de Frisbee, direcionado para a extensão do cotovelo, coordenação do pulso, integração visuoespacial e capacidade de reação. Foi ainda integrado um Módulo de Reconhecimento de Emoções como prova de conceito, transmitindo o estado emocional do sobrevivente para o ambiente VR de outro utilizador para mitigar o isolamento social. Avaliado através de um estudo de utilizadores em duas fases (N = 38), o protótipo obteve uma pontuação média de 82/100 na SUS (Excelente), com 82% dos participantes a percecionarem a dificuldade adaptativa em ação, 98% a reportarem uma experiência motivante, e 68% a solicitarem espontaneamente continuar a jogar após a sessão estabelecendo uma base sólida para futura validação clínica longitudinal.

keywords

Virtual Reality, Serious Game, Personalization User-centered design, Stroke, Rehabilitation

abstract

Stroke is one of the leading causes of long-term disability worldwide, progressively eroding survivors' motor function, cognition, and quality of life. Conventional rehabilitation programs suffer from two critical limitations: monotony arising from repetitive therapeutic exercises, and a lack of individualized adaptation both of which undermine patient motivation and engagement. This dissertation presents Country Fair VR, a Virtual Reality serious game prototype for the Meta Quest 3, designed to support upper limb and cognitive rehabilitation of stroke survivors. The system is built around a closed-loop personalization framework featuring a rule-based Dynamic Difficulty Adjustment (DDA) engine implementing a hybrid model that combines autonomous real-time adaptation with therapist manual override a preference validated by 70% of healthcare professionals surveyed at the Rovisco Pais Rehabilitation Center. The prototype features two therapeutically grounded mini-games set in a vibrant country fair environment: an Archery game targeting bimanual strength, proprioceptive training, and selective attention and a Frisbee game targeting elbow extension, wrist coordination, visuospatial integration, and reaction capability. An Emotion Recognition Module was additionally integrated as a proof-of-concept, relaying survivors' emotional states to peers' VR environments to mitigate social isolation. Evaluated through a structured two-wave user study (N = 38), the prototype achieved a mean SUS score of 82/100 (Excellent), with 82% of participants perceiving adaptive difficulty in action, 98% reporting a motivating experience, and 68% spontaneously requesting to continue playing after the session ended establishing a solid foundation for future longitudinal clinical validation.

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Glossary

AI	Artificial Intelligence	MT	Mirror Therapy
CIMT	Constraint-Induced Movement Therapy	NPCs	Non-Playable Characters
DDA	Dynamic Difficulty Adjustment	RL	Reinforcement Learning
FSS	Flow State Scale	SDK	Software Development Kit
HCD	Human-Centered Design	SUS	System Usability Scale
HMD	Head-Mounted Display	VAR Lab	Virtual and Augmented Reality Research Group
HUD	Heads-Up Display	VR	Virtual Reality
IEETA	Institute of Electronics and Informatics Engineering of Aveiro	XR	Extended Reality
ML	Machine Learning		

Introduction

Stroke is a leading cause of death and long-term disability worldwide, with over **12 million** new cases expected annually and more than **94 million** people currently living with its consequences. Survivors frequently face permanent impairments affecting mobility, cognition, speech, and vision that diminish their ability to perform daily activities and significantly reduce their quality of life, often leading to depression and social isolation [1] [2].

While survivors may not fully return to their pre-stroke state, it is recommended that rehabilitation begin as soon as possible to maximize the recovery of lost capacities and improve performance in daily activities. Rehabilitation plays a central role in addressing these challenges, particularly in restoring physical capacities such as motor function and mobility and cognitive abilities, including attention, memory, and executive function. Despite their proven benefits, conventional rehabilitation methods still present constraints that limit further adherence by stroke survivors. To better understand these limitations, consider the rehabilitation process as analogous to building a brick house, where the structure represents the recovery of the survivor's capacities. Just as a house cannot be built without bricks and mortar, successful rehabilitation requires fundamental components. In this analogy, the survivor's spirit, motivation, determination and engagement are the bricks the essential, foundational elements without them no progress can be made. The exercises constitute the mortar, binding the components together and translating motivation into visible progress. However, traditional rehabilitation often relies on the repetitive practice of the same exercises. This can make them feel tedious, monotonous over time, deterring the patient's motivation and engagement the essential '*bricks*' thereby compromising the entire recovery process [2] [3] [4] [5].

Another major constraint of conventional rehabilitation programs is the fact that they usually apply the same approach to everyone, neglecting the individual characteristics and recovery paths of each stroke survivor. As a result, survivors who struggle with the tasks while seeing their peers succeed may become demotivated, which again breaks the essential

'bricks' of this process. The core issue, therefore, does not lie with the survivors themselves, but rather with programs that fail to adapt to the specific needs and recovery pace of each individual [2] [3] [4] [5] [6] [7] [8].

This leads to an important question, **how can we create rehabilitation programs that adapt to the individual needs of each survivor?** The solution lies in introducing a certain degree of personalization into these programs focusing on adapting to the survivor's specific needs and progress.

However, monitoring a survivor's progress with the accurate precision and adapting the exercises in real-time based on their performance is overly complicated for healthcare professionals. Manually recording times, comparing results across sessions, and analyzing various performance parameters is time-consuming and inclined to error. Therefore, there is a clear need for tools that can automatically make these adjustments and assist professionals in monitoring progress.

When considering automation, digital solutions such as computers are what comes to mind. A bright approach is to consider the use of **Virtual Reality (VR)** combined with **Serious Games**. Digital environments exploring **VR** are well known for creating an immersing experience that increases user engagement. In addition, **VR** Headsets can track users hands/arms, allowing users to control **VR** applications in a natural way. This feature can be useful for stroke survivors practicing specific movements required to recover their lost motor skills. Serious Games are digital games which are designed for purposes beyond entertainment, but they incorporate gamification elements similarly or equal to those found on traditional digital games, which are responsible for keeping players motivated and engaged. Another similarity between this type of games and traditional ones is that they also have the benefit of improving cognitive functions. Serious Games are particularly known to be able to adapt to user's performance through automatic algorithms such as **Dynamic Difficulty Adjustment (DDA)** ensuring that the challenge still remains in the new level of difficulty and still maintaining the fun of playing the game. When combined these technologies have the potential to boost user motivation and engagement. However, despite broadly positive clinician attitudes toward **VR**, the vast majority of physiotherapists have yet to integrate it into clinical practice, underscoring the urgency of developing systems that are both clinically transparent and demonstrably usable [3] [9].

Despite the promising potential of **VR** and Serious Games for stroke rehabilitation, their adoption in clinical practice remains limited. A recent cross-sectional survey found that, while physiotherapists generally hold positive attitudes toward immersive **VR**, the vast majority have yet to integrate it into routine care, citing barriers related to cost, usability, and lack of clinical evidence. This gap between demonstrated potential and real-world adoption highlights the need for systems that are not only therapeutically grounded, but also clinically transparent and demonstrably usable. This dissertation addresses precisely this opportunity [3] [9].

1.1 GOALS AND CONTRIBUTIONS

This dissertation aims to design and develop a Personalized **VR** based Serious Game for physical rehabilitation that trains the upper limbs and cognitive functions of stroke survivors. The upper limb aspects that the games will focus on include arm and hand movements, gripping strength, and motor coordination, while the cognitive functions addressed are planning, attention, and reasoning.

To this end, the following objectives were defined:

- Develop a personalized **VR** serious game for stroke rehabilitation;
- Implement an automatic rule-based algorithm;
- Apply an **Human-Centered Design (HCD)** methodology;
- Evaluate usability, motivation, and personalization perceptibility.

To achieve this aim, the proposed serious game will need to offer the same initial difficulty for all survivors, progressively adapting to each individual's performance in real time. To accomplish this, the existing literature must be analyzed to understand which personalization and difficulty adjustment methods have already been explored and which can be effectively integrated into the proposed concept. To accomplish the defined objectives, a **HCD** methodology was adopted, involving a multidisciplinary team that included healthcare professionals from the **Rovisco Pais Rehabilitation Center**, to understand the needs, motivations, and expectations of the target population stroke survivors and healthcare professionals. This process informed the definition of requirements and design ideas for the proposed system. All in all, this thesis aims to highlight the potential of personalization in rehabilitation and to reinforce the idea that **VR** and serious games can serve as valuable complements to traditional rehabilitation programs, increasing patients' motivation and progress.

The contributions of this thesis are as follows:

- Analysis of existing literature and selection of a personalization method suited to the stroke rehabilitation context;
- Design and development of a **VR** based framework incorporating two serious games for the personalization of the virtual rehabilitation environment;
- Proposal of a Country Fair approach to stroke rehabilitation, combining immersive environments, adaptive difficulty, and emotion recognition;
- User evaluation conducted in two distinct moments with over **38** participants, encompassing healthcare professionals and general users;
- Integration of an emotion recognition module as input data for the personalization system, with shared emotional awareness displayed within the **VR** environment.

The results of this work have contributed to the following publications:

- Figueiredo, I., Palricas, D., Marques, B., Guimarães, A. R., Oliveira, S., Amorim, P., ... & Santos, B. S. (2026); Addressing Personalized Virtual Reality for Stroke Rehabilitation

Through a Country Fair Serious Game Approach: A Proposal; In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 914–920);

- Figueiredo, I., Marques, B., Palricas, D., Oliveira, S., Guerreiro, B., Reisinho, P., Silva, S., Dias, S., Amorim, P., & Santos, B. S. (2026). Comparing Methods for Integrating Personalized eXtended Reality in Stroke Rehabilitation. In *EUSSET Conference on Computer-Supported Cooperative Work (ECSCW)* (pp. 1–10);
- B.Guerreiro, D. Palricas, I. Figueiredo, S. Oliveira, B. Marques, S. Brás, P. Amorim and B. S. Santos (2026). "Visualization Framework for Shared Emotional Awareness in Personalized VR for Stroke Recovery". Proceedings of the International Conference on Information Visualisation, IV, 1-6.

1.1.1 Thesis Structure

This thesis is organized into five chapters. Chapter 1 introduces the problem context, motivation, research goals, and contributions of the present work.

Chapter 2 presents a review of the relevant literature, examining the clinical context of stroke rehabilitation, the role of **VR** and Serious Games in rehabilitation settings, personalization strategies and dynamic difficulty adjustment methods, as well as the application of **HCD** in the development of rehabilitation systems.

Chapter 3 describes the design and development of **Country Fair VR**, detailing the conceptual framework, the two serious games developed and the personalization module.

Chapter 4 presents the two waves of user evaluation conducted, describing the methodology, participants, procedures, and results obtained from both studies.

Chapter 5 concludes the thesis by summarizing the main findings and contributions, discussing the limitations of the present work, and outlining directions for future research.

Background and related work

This chapter presents an analysis of key concepts associated with the themes explored in this thesis, namely stroke and its impact on survivors, the role of **Virtual Reality (VR)** and Serious Games in rehabilitation contexts, personalization strategies and adaptive systems, and the application of Human-Centered Design in the development of rehabilitation technologies.

It is structured as a narrative review rather than a formal systematic one, with these domains selected for their direct relevance to the research questions established in Chapter 1. The technological core of this review, particularly the treatment of Extended Reality (XR), personalization, and difficulty adaptation, is anchored on the scoping review conducted by Figueiredo et al., which followed the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)** guidelines and surveyed the **SCOPUS** database over the **2020–2024** period, narrowing **82** retrieved records to **39** studies through predefined inclusion criteria and a search query spanning eXtended Reality, personalization/adaptation, rehabilitation, patient, and stroke terms. The foundational clinical and theoretical concepts that lie outside the technological scope of that review, such as stroke epidemiology, neuroplasticity, the evidence base for established motor rehabilitation therapies, and **Flow Theory**, were drawn from targeted searches of primary studies, meta-analyses, and seminal works, ensuring that each thematic area rests on sources appropriate to its nature. [3]

2.1 STROKE CAUSES AND CHALLENGES

Stroke is defined as an acute episode of focal dysfunction of the brain, retina, or spinal cord lasting longer than 24 hours or resulting in death, and is broadly classified into ischemic (arterial occlusion) and hemorrhagic (intracerebral or subarachnoid bleeding) subtypes. According to the **World Stroke Organization**, over 100 million people have already experienced a stroke, more than 94 million people globally live with its consequences, and over 15 million new cases occur annually [1] [10] ².

¹Source: Living Healthy Hawaii stroke risk infographic.

²World Health Organization. *Stroke/Cerebrovascular accident*. 2025. Available at: <https://www.emro.who.int/health-topics/stroke-cerebrovascular-accident/> [Accessed: 15 January 2026]

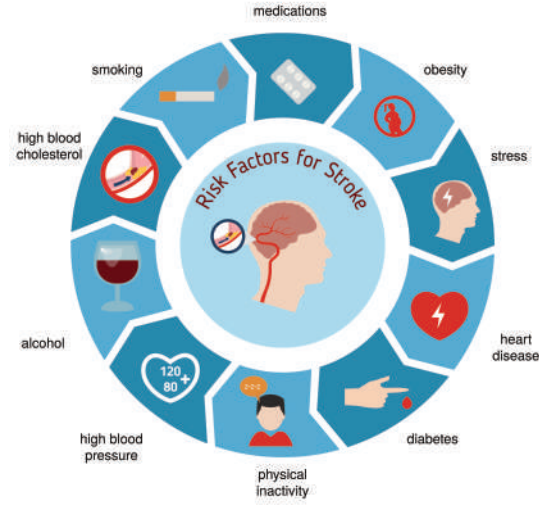


Figure 2.1: Overview of the principal modifiable risk factors associated with stroke, including metabolic conditions (obesity, diabetes, and hypercholesterolaemia), behavioural factors (physical inactivity, smoking, and heavy alcohol use).¹

Modifiable risk factors account for the large majority of stroke burden: **91%** of stroke risk is attributable to metabolic factors such as obesity, hyperglycemia, and hyperlipidemia, **74%** to behavioral factors including smoking, physical inactivity, and poor diet, and **29%** to air pollution. As illustrated in Figure 2.1, these risk factors span a broad range of metabolic, behavioral, and cardiovascular domains, and the resulting impairments affect motor, cognitive, and communication function [2] [11].

A critical but often underemphasized dimension of stroke recovery is **neuroplasticity** the brain’s capacity to reorganize its structure and function in response to experience and activity. Neuroplasticity underpins all motor rehabilitation and is maximized when practice is frequent, task-specific, and sufficiently challenging. Recovery is not uniform over time: clinically, stroke rehabilitation is divided into three phases with distinct biological and therapeutic profiles. The **acute phase** (under 2 weeks post-stroke) prioritizes medical stabilization and early mobilization. The **subacute phase** (2 weeks to 6 months) represents the window of highest neuroplastic potential, during which intensive motor retraining yields the greatest functional gains. The **chronic phase** (beyond 6 months) is characterized by slower recovery, though meaningful improvement remains achievable through sustained practice. The system developed in this work targets survivors in the **subacute and chronic phases**, where engagement and personalization are most critical to sustaining the intensity of practice required for neuroplastic change [6] [7].

Early initiation of rehabilitation is consistently associated with better functional outcomes, as it capitalizes on the neuroplastic potential of the subacute phase. Among the most evidence-supported approaches for upper limb motor recovery are **Constraint-Induced Movement Therapy (CIMT)** and **Mirror Therapy (MT)** [2].

Through the constraint of the non-paretic limb, **CIMT** enforces intensive use of the affected arm through repetitive, task-oriented practice. A systematic review and meta-analysis

by Kwakkel et al. (2015) confirmed that **CIMT** produces statistically significant improvements in arm motor function and activity performance compared to conventional therapy, with effect sizes that are clinically meaningful across both subacute and chronic populations. By placing a mirror in the midsagittal plane, **MT** creates a visual illusion of movement in the paretic limb, engaging mirror neuron systems to facilitate motor relearning. A Cochrane review by Thieme et al. (2018) analyzed 62 trials and found moderate-quality evidence that **MT** improves motor function, activities of daily living, and pain in post-stroke patients compared to control interventions [12] [13].

Despite their documented efficacy, both methods share a fundamental structural limitation: they rely on the systematic repetition of a fixed exercise regime. This inherent monotony is associated with declining motivation and adherence over the extended recovery periods typical of stroke rehabilitation, particularly in the chronic phase where external clinical supervision is reduced. The lack of adaptive challenge exercises that respond to the patient’s evolving performance further compounds disengagement, as tasks that are too easy or too repetitive fail to sustain the level of active attention required for neuroplastic change. Rehabilitation through immersive virtual environments directly addresses both of these limitations by embedding therapeutic motor tasks within interactive, gamified environments that adapt to the patient’s performance, sustaining engagement across repeated sessions [3] [5] [6] [14].

2.2 DIGITAL REALITIES AND SERIOUS GAMES IN STROKE REHABILITATION

XR is an umbrella term encompassing technologies that alter perceived reality by integrating digital elements into physical or virtual environments, including **VR**, which immerses the user in a fully digital environment, **Augmented Reality (AR)**, which overlays digital content onto the real world, and **Mixed Reality (MR)**, which blends physical and virtual elements that can interact in real time (Figure 2.2). These technologies have been considered responses to the core limitations of traditional stroke rehabilitation: monotonous exercise routines and declining patient motivation [15] [16] [17] [18] [19].

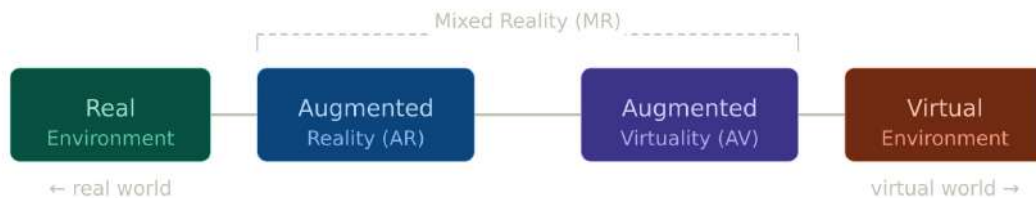


Figure 2.2: Reality Virtuality Continuum, adapted from [15].

A scoping review by Figueiredo et al. (2025) of **39 publications** retrieved from the **SCOPUS** database (**2020-2024**) confirms that **VR** is the dominant modality, appearing in **84.6%** of the analyzed studies. This dominance is attributed to **VR**’s capacity to generate

fully immersive, controlled environments in which patients can practice motor and cognitive tasks without the risks of physical settings [3] [20].

Serious Games constitute the dominant intervention strategy, appearing in **67%** of the reviewed exercises. Formally defined as games designed for a primary purpose other than pure entertainment encompassing education, training, and health serious games embed game mechanics to engage users in the acquisition of skills or behaviors through interactive scenarios. A prominent subtype is the **exergame** (a portmanteau of *exercise* and *game*): a serious game in which bodily movement is the primary input mechanism, and the physical exertion required to play is itself the therapeutic exercise. In stroke rehabilitation, this translates to converting repetitive therapeutic movements into structured gameplay: from gesture-based rhythm games targeting hand dexterity, to rule-based exergames for upper limb motor recovery, to adaptive systems that infer affective states from movement patterns to adjust session difficulty in real time [3] [21] [22] [23] [24] [25].

From a clinical perspective, **upper limb rehabilitation** is the primary target, addressed by **66.7%** of the reviewed studies a proportion that directly reflects the high prevalence of post-stroke hemiparesis and its disproportionate impact on daily autonomy. Regarding interaction modality, hand-tracking is used as the primary interface in **41%** of studies, outperforming physical controllers (**25.6%**). This shift is clinically motivated: grasping a controller presents a significant barrier for survivors with limited hand function, whereas hand-tracking enables natural interaction without requiring grip strength or fine motor precision, as demonstrated by Bae and Park, whose rhythm-based serious game used gesture recognition as the sole interaction modality and reported measurable improvements in hand rehabilitation outcomes [3] [21].

Beyond engagement, a subset of these studies reports clinically relevant functional outcomes. Villada Castillo et al. (2024) conducted a usability study with a **VR** exergame for upper limb stroke rehabilitation and reported improvements across motor performance metrics alongside positive usability ratings, demonstrating that serious game design and clinical efficacy are not mutually exclusive. Ferreira and Menezes (2020) developed one of the few systems targeting physical and cognitive rehabilitation simultaneously within a single adaptive **VR** environment, a design choice directly relevant to the dual-domain rehabilitation goals of the present work. Rivas et al. (2020) demonstrated that affective states including frustration and fatigue can be inferred unobtrusively from upper limb motion during **VR** rehabilitation, providing an evidence base for emotion-aware adaptation mechanisms beyond biomechanical metrics [22] [26] [27].

Complementing serious game approaches, **23%** of the reviewed studies use simulations of real-life therapeutic activities (e.g., mirror therapy, palmar grasping exercises), and **10%** simulate Activities of Daily Living such as shopping or household tasks, prioritizing the recovery of functional autonomy. The addition of gamification elements to these platforms has demonstrably reduced the monotony inherent in repetitive physical therapy [3] [5].

2.3 PERSONALIZATION STRATEGIES IN VR REHABILITATION

Personalization is identified as a key element of effective therapy due to the recognition that each stroke survivor’s course of recovery is unique and varies according to factors like physical, cognitive, and emotional impairments. Mainstream approaches to this issue through rehabilitation also offer standard or “one-size-fits-all” methods that do not adequately address individual needs. The inadequacy of these processes is also established to hinder recovery due to factors like frustration and lack of motivation on the part of survivors [3].

It is worth noting that personalization is one facet of the broader field of **game adaptivity**, surveyed by Lopes and Bidarra (2011), which frames the dynamic adjustment of game components, such as game worlds, mechanics, non-playing characters, and scenarios, as a process steered by **player modelling** and realized through either **offline** methods, which generate customized content ahead of play, or **online** methods, which adjust the experience in real time. Framing the present work through this lens situates its real-time difficulty adjustment within the online branch of a wider design space, while leaving the largely unexplored offline, cross-session dimension as a direction for future work [28].

2.3.1 Taxonomy of Personalization

Current research has offered a taxonomy to structure how **XR** environments can be adapted to individual needs. The review mentioned before (Figure 2.3) identified five key personalization approaches, distributed as follows [3]:

- **Behavioral Tracking (30.8%)**: Monitors movement, posture, and engagement patterns to infer states such as frustration or fatigue, triggering adaptive responses in real time;
- **Rule-based Systems (25.6%)**: Apply predefined logical rules (e.g., “if the patient fails three consecutive attempts, reduce difficulty”) to drive adaptation; straightforward to implement and clinically interpretable;
- **Bio-Feedback (23%)**: Use physiological signals such as heart rate, **electromyography (EMG)**, or **electroencephalogram (EEG)** to drive real-time adaptation based on the patient’s measured physical or cognitive state;
- **Contextual Adaptation (21%)**: Adjusts the session based on external variables such as time of day, session history, or inferred patient mood, enabling session-level rather than moment-to-moment personalization;
- **Intelligent Monitoring (0%)**: Employs **Machine Learning (ML)** algorithms to analyze longitudinal data patterns and predict user needs proactively. Despite its theoretical potential, no study in the reviewed corpus implemented this approach. The scoping review attributes this gap to the substantial requirements of **ML** models large labeled datasets, real-time processing infrastructure, and algorithmic transparency combined with the financial constraints of rehabilitation centers and legitimate concerns about patient data privacy [3].

Each of these approaches presents distinct trade-offs that have direct implications for system design. **Rule-based systems** are interpretable and safe to deploy clinically, but

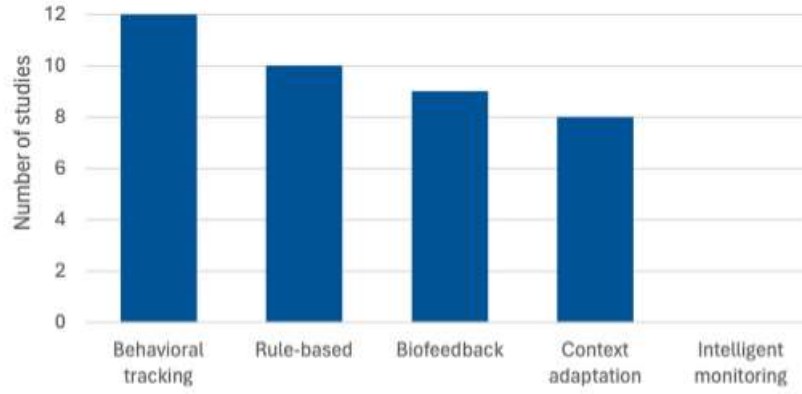


Figure 2.3: Distribution of publications across different personalization methods, highlighting Behavioral tracking as the most studied approach, followed by Rule-based and Biofeedback methods. Adapted from [3]

their static thresholds cannot capture the non-linear and fluctuating nature of stroke recovery a patient who fails three times due to transient fatigue is not the same as one failing due to insufficient skill progression. **Behavioral tracking** offers greater responsiveness but depends on the validity of proxies: inferred frustration from movement data is not equivalent to measured frustration, and misclassification can produce maladaptive difficulty changes. **Bio-feedback** provides the most direct physiological grounding, but its reliance on additional sensors increases setup complexity and patient burden a significant barrier in populations with motor and cognitive impairments. **Contextual adaptation** operates at a coarser temporal resolution, making it better suited for between-session adjustments than real-time responsiveness. Critically, none of these approaches as currently implemented maintains a persistent model of the patient’s longitudinal trajectory the gap that Intelligent Monitoring is designed to address, and that justifies the hybrid heuristic-clustering design adopted in the present work (Section 2.4) [29].

2.3.2 Automation vs Clinical Control

One major issue associated with personalization is methods and styles of adaptation, and more generally management or orchestration. While it appears that there is a preponderance towards automatic personalization (**46.2%**) in existing literature, this does not necessarily reflect professional preference [3].

A recent study show a dominant leaning toward the hybrid model (**70%**)(Figure 2.4). The preference is for systems that can either suggest or execute the changes through software but still enable the therapist to correct the system with their own interventions whenever deemed necessary. The system designed for human involvement allows for oversight and reduces the therapist’s burden of working manually. Additionally, concerning the point of executing the modifications, **50%** of healthcare professionals lean toward the Hybrid model of synchronization, which aims to modify the system both during the actual process and subsequently using historical data [4].

With respect to the data used as inputs for the personalization process, although traditional

measures of physical activity (such as range of movement) were common, personalized therapy requirements for the survivor were seen as motivation levels (**70%**) and emotional state (**60%**) by health professionals. This is an indication that the survivor’s psychology has been targeted for enhanced therapy compliance [4].

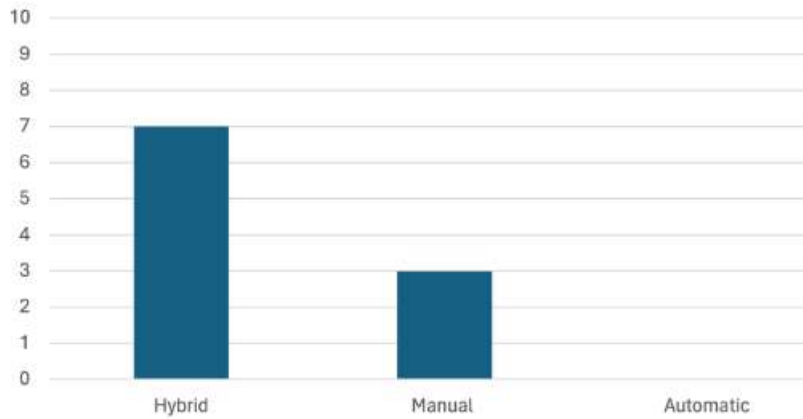


Figure 2.4: Distribution of healthcare professionals votes for each personalization mode: hybrid (7), manual (3), automatic (0).Adapted from [4]

2.4 DYNAMIC DIFFICULTY ADJUSTMENT (DDA) AND FLOW THEORY

One of the central challenges of stroke rehabilitation is sustaining the survivor’s active engagement across extended periods of repetitive exercise. A theoretically grounded response to this challenge is found in **Flow Theory**, formally proposed by Csikszentmihalyi in his seminal work *Flow: The Psychology of Optimal Experience*. According to this framework (Figure 2.5), optimal experience, the **Flow Zone** occurs when the perceived challenge of a task is in equilibrium with the individual’s skill level. When challenge exceeds skill, the result is anxiety and frustration; when skill exceeds challenge, the result is boredom and disengagement. Both conditions are detrimental to rehabilitation adherence [8] [30] [31].

The operational validity of **Flow Theory** in interactive contexts has been established empirically. Jackson and Marsh developed and validated the **Flow State Scale (FSS)**, a nine-dimension psychometric instrument that measures flow across dimensions including challenge skill balance, absorption, and sense of control. The **FSS** has since been applied to serious game contexts, confirming that well-calibrated adaptive difficulty is a significant predictor of flow state and sustained engagement [31] [32].

While Csikszentmihalyi’s model provides the foundational account of optimal experience, subsequent work has extended it to better fit the dual objectives of rehabilitation exergames. The **Dual Flow Model** (Figure 2.6) proposed by Sinclair et al. decouples the classic *attractiveness* axis balancing challenge and skill to sustain engagement from an orthogonal *effectiveness* axis, which ensures that the physical intensity of play remains within a therapeutically beneficial range. This separation is directly relevant to the present work, in which a

task must be simultaneously engaging and clinically effective: a patient may remain in flow while exercising at an intensity that is therapeutically insufficient, a condition that single-axis flow models cannot capture [33].

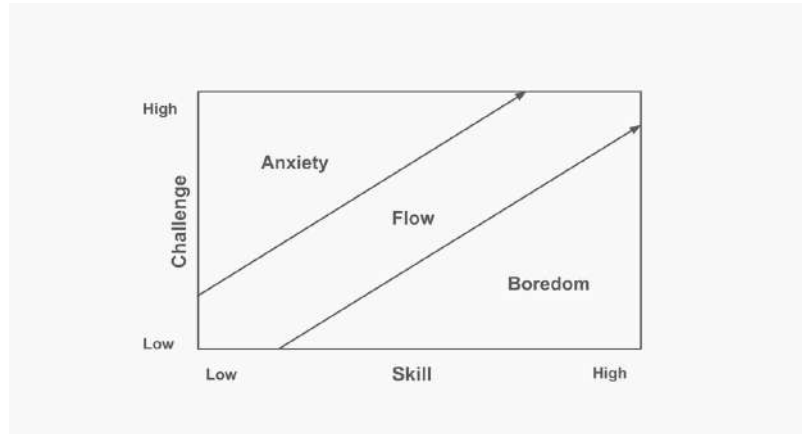


Figure 2.5: Mihaly Csikszentmihalyi’s Flow Theory: the Flow Zone is reached when challenge and skill are in balance. Too much challenge produces anxiety, too little produces boredom. Adapted from [30].

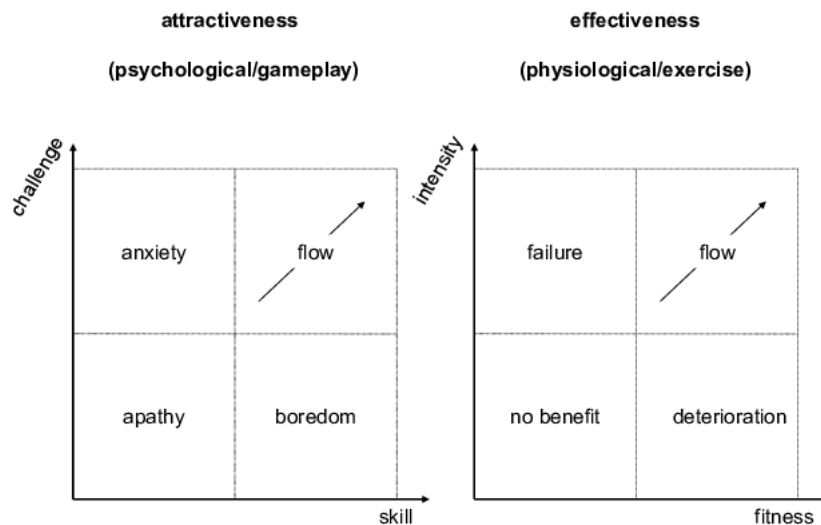


Figure 2.6: The Dual Flow Model for exergaming proposed by Sinclair et al. It extends Csikszentmihalyi’s single-axis flow with two orthogonal dimensions: an *attractiveness* axis, balancing challenge against skill to sustain engagement, and an *effectiveness* axis, ensuring that the physical intensity of play remains within a therapeutically beneficial range. Adapted from [33].

2.4.1 Mechanisms of Adaptation

To operationalize **Flow Theory** within **VR** rehabilitation, systems employ **DDA** defined as the capacity of a system to modify task parameters in real time in response to the player’s

measured performance. The goal is to continuously position the patient within the Flow Zone, preventing both anxiety from excessive difficulty and disengagement from insufficient challenge [8].

In the literature, **DDA** approaches span a spectrum of computational complexity. At one end, **heuristic methods** apply fixed logical rules (e.g., “if the patient fails three consecutive attempts, reduce difficulty by one level”) that are transparent, easy to validate clinically, and require no training data. At the other end, **Reinforcement Learning (RL)** frames difficulty adaptation as a sequential decision problem, where an agent learns a policy that maximizes long-term engagement by exploring difficulty adjustments and observing their effect on patient behavior. **Bayesian approaches** offer a middle ground, maintaining a probabilistic model of the patient’s latent skill state that is updated after each observation, allowing principled uncertainty quantification and personalized adaptation even with limited session data. Despite their theoretical advantages, both **RL** and **Bayesian methods** present significant barriers in clinical rehabilitation contexts: **RL** requires large interaction histories to converge to stable policies, which are rarely available for stroke populations with heterogeneous recovery trajectories, **Bayesian methods** require careful prior specification and are difficult to audit or explain to clinical staff.

A further, data-driven route to personalization is the profiling of survivors through the clustering of their performance data. Bouatrous et al., for instance, applied an unsupervised K-means algorithm to group patients according to their measured performance during a **VR** hand-rehabilitation exergame, using the resulting skill-level clusters to drive real-time difficulty adaptation. This behaviour-based grouping is conceptually analogous to player profiling, but is grounded in measured motor performance rather than predefined player types, offering an interpretable middle ground between static rule-based thresholds and the opaque, data-intensive reinforcement-learning approaches discussed above [20].

These interpretability and data requirements directly informed the design choice in the present work, described next [3] [31] [34] [35].

2.5 HUMAN-CENTERED DESIGN AND CLINICAL ACCEPTANCE

The effectiveness of innovative rehabilitation technologies is not contingent solely on their technical performance, but equally on how well they integrate into clinical workflows and align with the needs of all stakeholders. Despite growing evidence for the therapeutic potential of **VR**, its adoption in clinical practice remains limited. Scoping reviews consistently identify a persistent gap between clinicians’ generally positive attitudes toward **VR** and its actual integration into routine care, attributing this gap to a complex interplay of organizational, technological, and clinician-related barriers. This reality underscores why the latest generation of rehabilitation systems increasingly adopts a **Human-Centered Design (HCD)** approach, which engages end-users both stroke patients and healthcare professionals throughout the entire development process [36] [37].

2.5.1 Bridging the Gap: Role of Health Care Professionals

Healthcare professionals act as gatekeepers of rehabilitation interventions: they prescribe, monitor, and ultimately determine whether a technology is integrated into patient care. Research consistently shows that clinician-related factors including perceived ease of use, self-efficacy, and alignment with existing clinical protocols are among the most critical determinants of **VR** adoption in rehabilitation settings. A cross-sectional survey of French- and Dutch-speaking physiotherapists across eight countries found that **75.5%** had never used immersive **VR** in clinical practice, despite most holding positive attitudes toward the technology and perceiving it as clinically useful. The primary barriers cited equipment cost, lack of institutional support, and insufficient training time are organizational in nature, not attitudinal, suggesting that acceptance alone is insufficient to drive adoption [9] [36] [37].

This disconnect between acceptance and adoption is further evidenced at the local level. A study conducted at the *Centro de Reabilitação Rovisco Pais* found that existing technological solutions frequently overlook the realities of clinical practice, failing to accommodate the workflow and customization needs of therapists. Taken together, these findings indicate that successful **VR** rehabilitation systems must be designed not only for patients, but with and for the clinicians who deliver care [4].

The **HCD** framework also highlights a paradigm shift in what data matters most for personalization. Traditional rehabilitation systems prioritize biomechanical metrics such as range of motion and movement speed. However, when healthcare professionals were asked to identify the most critical data types for tailoring recovery, they cited Motivation (**70%**) and Emotional State (**60%**) above physical performance indicators. This finding reinforces the need for **VR** systems that go beyond motor training, integrating gamification and engagement-monitoring features capable of detecting and responding to psychological states a design requirement directly addressed in the present work [4].

2.6 SUMMARY

The review conducted in this chapter identifies a consistent set of gaps across the clinical, technological, and design dimensions of **VR** stroke rehabilitation. Rather than recapitulating each section independently, Table 2.1 presents these gaps in structured form, mapping each identified limitation directly to the design decision it motivates and to the chapter where that decision is operationalized.

These converging gaps collectively define the design space for the present work. The target interaction modality is **hand-tracking** the personalization architecture is **hybrid**, preserving clinical authority while automating routine difficulty adjustments, the adaptation mechanism is **rule-based** and grounded in **Flow Theory** and the therapeutic content targets **upper limb motor function** within an ecologically valid gamified environment.

Table 2.1: Summary of literature gaps and resulting design decisions.

Gap identified	Design decision
Traditional rehabilitation (CIMT , MT) relies on monotonous repetition, reducing adherence in the subacute and chronic phases where sustained practice is most needed.	Embed therapeutic motor tasks within a gamified VR serious game (<i>Country Fair VR</i>) to sustain engagement across repeated sessions.
Physical controllers exclude stroke survivors with severe upper limb impairment, hand-tracking dominates in 41% of the reviewed literature but remains underexplored for archery-based interaction.	Adopt hand-tracking as the sole interaction modality, removing the grip requirement and enabling natural gesture-based gameplay.
Existing VR rehabilitation systems adopt fully automated personalization, which conflicts with the clinical preference for therapist oversight reported by 70% of healthcare professionals.	Implement a hybrid personalization model: rule-based DDA handles real-time difficulty adjustment; the therapist retains override authority via a dedicated interface.
DDA approaches in the literature prioritize biomechanical metrics; neither engagement nor emotional state is used as an adaptation signal, despite clinicians rating motivation (70%) and emotional state (60%) as the most critical personalization inputs.	Integrate gamification mechanics reward systems, progressive challenges, environmental feedback designed to maintain the patient within the Flow Zone as defined by <i>Csikszentmihalyi</i> [30].
Advanced ML -based DDA approaches (RL , Bayesian methods) require large datasets and produce non-interpretable policies, making them unsuitable for clinical deployment with heterogeneous stroke populations.	Adopt a rule-based adaptation mechanism: performance thresholds trigger discrete difficulty changes, ensuring clinical transparency and immediate auditability by the therapist.
Only 30.8% of reviewed systems provide access to previous session data; longitudinal progress tracking essential for adaptive rehabilitation remains largely absent.	Implement a session history module that logs performance metrics across sessions, enabling both therapist review and data-informed difficulty seeding at session start.

Personalized VR-based Country Fair

This chapter presents the design and implementation of **Country Fair VR**, a **Virtual Reality (VR)** serious game prototype developed for stroke rehabilitation. Following the **Human-Centered Design (HCD)** methodology, it covers the system architecture, the Country Fair environment and its integration with the personalization model, the therapeutic justification and interaction mechanics of each mini-game, the *CarneyWise* personalization module, and the emotion recognition module incorporated as a complementary increment to the system.

3.1 METHODOLOGY

The development of **Country Fair VR** followed a **HCD** approach, consistent with the methodology adopted by the research group in which this work is integrated, as illustrated in Figure ???. The process began with visits to the **Rovisco Pais Rehabilitation Center** (Figure 3.2), which provided direct insight into the clinical context and the needs of stroke survivors and healthcare professionals. These findings, combined with the collective knowledge of the research group, informed the definition of the **Country Fair** concept and the selection of **hand-tracking** as the primary interaction modality a choice grounded in the group's experience that controller-free interaction is better suited to the motor constraints of the target population. The primary objective driving all design decisions is the **personalization of the rehabilitation experience during the execution of the activities**, adapting in real time to each survivor's individual performance and skill level. This focus on in-session personalization distinguishes this work from approaches that rely solely on pre-session configuration by healthcare professionals. The concept was subsequently refined through iterative development and two waves of user testing, as detailed in Chapter 4 [4] [5].



Figure 3.1: HCD methodology adopted for the design and development of the personalized VR serious game, structured into iterative phases spanning requirements extraction with stroke survivors and healthcare professionals, prototyping, and waves of user testing.



Figure 3.2: Illustration of the research team conducting a user study, with a stroke survivor answering a post-task questionnaire, with the support of the research members.

3.2 ARCHITECTURE

The system architecture of **Country Fair VR**, represented in Figure 3.3, was grounded in the conceptual framework proposed (Figure 3.4), which defines a closed-loop rehabilitation model integrating a **VR environment**, a **Data Hub**, a **Personalization Engine**, and a **Monitoring Tool**. The implemented architecture operationalizes this model through three concrete modules: a **Game** module, a **Personalization** module, and an **Emotion Recognition** module. The **Data Hub** responsible for aggregating session data and comparing current performance against historical records to inform adaptation decisions and the

Monitoring Tool a dedicated interface for healthcare professionals to supervise and adjust the rehabilitation process are part of the long-term vision for the framework but were not implemented within the scope of this dissertation [5].

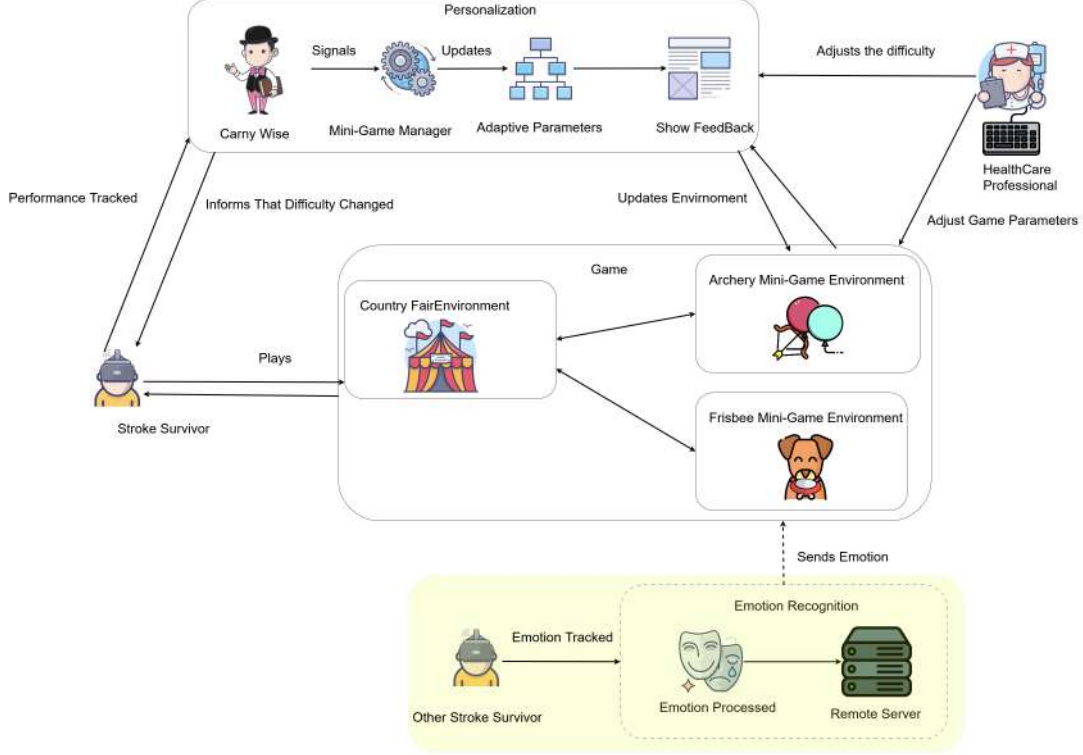


Figure 3.3: System architecture of **Country Fair VR**. The architecture contains three primary modules: (1) a **Personalization** module, managed by the *Carry Wise* character, which signals the Mini-Game Manager to dynamically adjust adaptive parameters and deliver multimodal contextual feedback; (2) a **Game** module, structured around a Country Fair Environment as the central hub, where the player can select between two distinct mini-game environments, Archery and Frisbee, whose configuration can be modified by the Personalization module through the adjustment of adaptive parameters. A *Healthcare Professional* interacts with the system to configure and monitor patient sessions, while the *Stroke Survivor* engages with the **VR** environment through a head-mounted display.

The **game module** is the main one of the system, which presents the initial scene, a vibrant “**Country Fair**” environment within the survivor’s **Head-Mounted Display (HMD)**. Here, players receive a brief introduction outlining what they will do in the game. Once the introduction is finished, they can select one of the available mini-games: the Frisbee or the Archery as illustrated in Figure 3.5

During a survivor’s first playthrough, both mini-games provide a short explanation of the rules alongside a practical tutorial. Within these mini-games, survivors must achieve a specific target, simulating a clinical session goal. Upon reaching this goal, they return to the country fair, where a congratulatory message is displayed to positively reinforce their performance. Throughout the gameplay, the mini-games’ difficulty dynamically changes based on the user’s performance, controlled by the **Personalization Module**. Additionally, players can return

to the country fair at any time by simply pressing a button located on their right side.

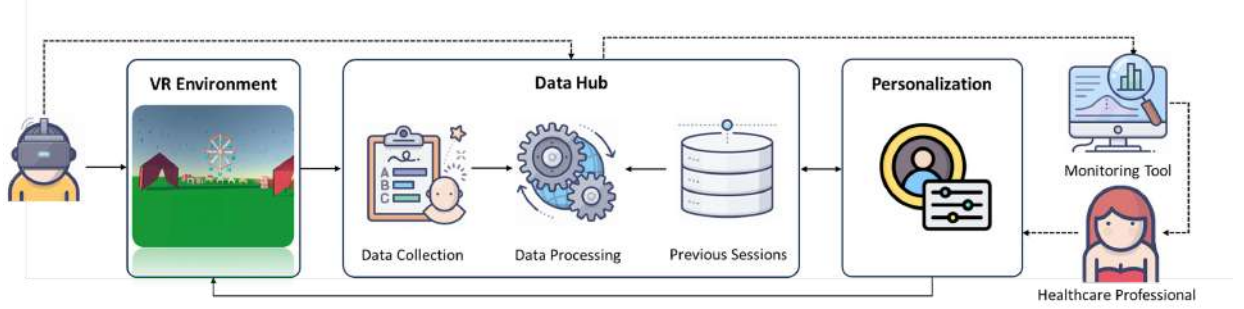


Figure 3.4: Conceptual framework for a closed-loop **VR** rehabilitation system combining **Dynamic Difficulty Adjustment (DDA)** and hybrid personalization to sustain motivation and promote neuroplasticity in stroke survivors. Adapted from [5].

Healthcare professionals can manipulate this module using a peripheral keyboard connected to the **HMD**. To support both development testing and clinical oversight, a set of keyboard shortcuts referred to as **cheatcodes** were implemented, allowing runtime manipulation of the game state without interrupting the session. Table 3.1 lists all available commands, their function, and the context in which they apply.

A detailed discussion regarding how the survivor interacts with the game, including the specific rules, goals, and mechanics of the mini-games, will be provided in Section 3.4.1.

The **personalization module** represents the main contribution of this dissertation and will be discussed in detail in Section 3.3. The **emotion recognition module** appears highlighted in Figure 3.3 not because it was a primary objective of this dissertation, but rather because it is a complementary integration of a colleague’s work (Bianca Guerreiro) to explore the potential of emotional awareness within the framework, as detailed in Section 3.5 [38].



Figure 3.5: The two mini-games integrated in the **Country Fair VR** system: the Archery mini-game (left), where the player uses a virtual bow to pop balloons, and the Frisbee mini-game (right), where the player throws a frisbee towards target areas for a virtual dog to retrieve.¹

VR Personalized Country Fair Demo: Click Where To Watch.

Table 3.1: Keyboard cheatcodes available to healthcare professionals during runtime.

Input	Function	Context	Precondition
intro	Skips the fair introduction sequence	Fair Scene	Intro not completed
archery	Loads the Archery mini-game	Fair Scene	Intro completed
frisbee	Loads the Frisbee mini-game	Fair Scene	Intro completed
return	Returns to the Fair Scene	Both mini-games	None
tutorial	Skips the mini-game tutorial sequence	Both mini-games	Tutorial not completed
increase	Signals <i>CarnyWise</i> to raise the difficulty by one level	Both mini-games	Tutorial completed
decrease	Signals <i>CarnyWise</i> to lower the difficulty by one level	Both mini-games	Tutorial completed
reset	Resets the difficulty to the first level	Both mini-games	Tutorial completed
e(expression_name)	Displays the specified emotion via the emoji representation	Both mini-games	Tutorial completed
positive/ negative	Shifts the emotion slider right or shifts it left, respectively, triggering the slider representation	Both mini-games	Tutorial completed
f(expression_name)	Displays the specified emotion via the facial expression representation	Both mini-games	Tutorial completed
score	Simulates a successful player score	Both mini-games	Tutorial completed
miss	Simulates a player miss	Both mini-games	Tutorial completed
dog	Simulates a score by landing the frisbee in the dog's surrounding area	Frisbee only	Tutorial completed

3.3 COUNTRY FAIR MODEL AND PERSONALIZATION INTEGRATION

The game is set in a “**Country Fair**”, an environment chosen for its cultural resonance, emotional value, and ability to promote a positive atmosphere. The fair is structured around a set of tents, each representing a distinct rehabilitation activity in the form of a serious game. To select a mini-game, the player looks directly at the corresponding tent, triggering a contextual prop representing the respective mini-game alongside a spatial 2D element displaying the mini-game’s title and an interaction button, which when pressed transports them to the selected activity. This tent based structure was conceived as a scalable system: while the scope of this dissertation covers two mini-games developed by the author, the model is designed to accommodate additional games in the future, including those already developed by other members of the research group, progressively expanding the range of rehabilitation activities available to survivors. The visual environment is designed to be vibrant and stimulating, incorporating dynamic structures, such as a rotating Ferris wheel, and ambient animals. Furthermore, to mitigate the feeling of social isolation often associated with rehabilitation therapy, the fair is populated with a crowd system of **Non-Playable Characters (NPCs)**, providing a crucial sense of social presence [4] [39].

Within this dissertation, two mini-games were developed and integrated into the system: an **Archery** game, where the player uses a virtual bow to pop balloons, and a **Frisbee** game, where the player throws a frisbee towards target areas for a virtual dog to retrieve, as illustrated in Figure 3.5.

To reinforce the carnival aesthetic and enhance visual engagement, a post-processing effect was applied to the virtual environment, elevating contrast and saturation levels. This stylistic choice contributes to a more immersive and emotionally resonant experience. Complementing the visual treatment, the **Heads-Up Display (HUD)** and the dialogue box of **NPCs** employs a custom carnival-style font (**Carnevalee FreakShow**), ensuring typographic consistency with the thematic setting and improving readability for users.

Upon entering the fair, the player receives a brief introduction given by the owner of the fair (**Zeca Bigodes**) (Figure 3.6). He explains the purpose of the game to the player, lists the mini-games that the fair offers, and introduces another character, **Carny Wise** (Figure 3.7), who will explain the rules of each mini-game, provide a practical tutorial for first-time players (including a demonstrative video showing the gesture to be performed), and ensure that the game matches the player’s skill level serving as the in-game representation of the personalization module. The name *CarnyWise* derives from the colloquial North American word “*Carny*”, referring to carnival workers, and “*Wise*”, indicating an intelligent character. As illustrated in Figure 3.3, this character controls a closed-loop rule based personalization system that continuously evaluates player performance to maintain a “**Flow State**” an optimal balance between skill and challenge that prevents both anxiety and boredom [8] [31].

The current mini-game available are: an Archery game, where the targets are balloons, and a Frisbee game, where the player must throw a frisbee to certain areas for a dog to catch in order to score points.

After the player finishes playing one of these games, they are transported back to the fair, and the owner congratulates them on his performance as shown in Figure 3.6.



Figure 3.6: Zeca Bigodes dialogue panels: introduction to the fair (top), positive performance feedback after completing the Frisbee mini-game (bottom left), and positive performance feedback after completing the Archery mini-game (bottom right).



Figure 3.7: Carny Wise tutorial sequence: instructional dialogue accompanied by a demonstrative video showing the gesture to be performed, during the Frisbee mini-game tutorial (left) and the Archery mini-game tutorial (right).

3.3.1 Personalization Hybrid Approach

According to the preferences expressed by healthcare professionals, the personalization module was designed as a **hybrid personalization model**. This architecture combines real-time algorithmic adjustments with the capability for human intervention, effectively dividing the system into a manual and an autonomous component [4].

Manual Component

Although initially implemented as developer shortcuts during the development phase, these cheatcodes (see Table 3.1) remain active in the final prototype build for the **HMD**. This functionality empowers healthcare professionals supervising the rehabilitation sessions to manually intervene and adjust the mini-games' difficulty based on their clinical judgment, provided a keyboard is connected to the **HMD**.

Autonomous Component

Beyond this manual override capability, the core of *CarnyWise* operates through an autonomous algorithm logic described in Figure 3.8 divided into two main branches:

- **Failure Handling:** If the player misses more than four consecutive times, the system immediately signals the mini-game manager to decrease the difficulty. This mechanism

was designed to automatically adjust the difficulty whenever the current level proves unsuitable for the player, including situations of potential fatigue, thereby preventing frustration. It should be noted, however, that the number of consecutive misses constitutes only an indirect and approximate indicator of fatigue, rather than a concrete measure of it. A more rigorous assessment of this phenomenon would require other methods, such as behavioral detection or biofeedback techniques, which are capable of providing more direct physiological or behavioral indicators of the player’s fatigue state [3];

- **Precision Analysis:** When the player scores, the system calculates the “Score Precision” percentage. This metric is calculated as the inverse of the number of attempts required to achieve a successful score (i.e., $P = \frac{1}{n}$, where n represents the total number of attempts). This branch relies on a heuristic, rule-based behavior system, which the literature identifies as the most common for stroke rehabilitation. Specifically, a high precision ($\geq 60\%$) increments an Excel Counter, while a low precision ($\leq 40\%$) increments a Struggle Counter. Reaching a predefined threshold in either counter triggers an automatic adjustment in the current mini-game’s difficulty [3].

It should be noted that the specific numeric values commanding these branches, both the 60%/40% precision bounds and the thresholds that trigger an adjustment, were set to provisional values during development for testing purposes, rather than being derived from a systematic empirical or clinical calibration. They were chosen pragmatically to validate the adaptation logic and are exposed as configurable parameters, so that the mechanism itself is decoupled from any particular tuning.

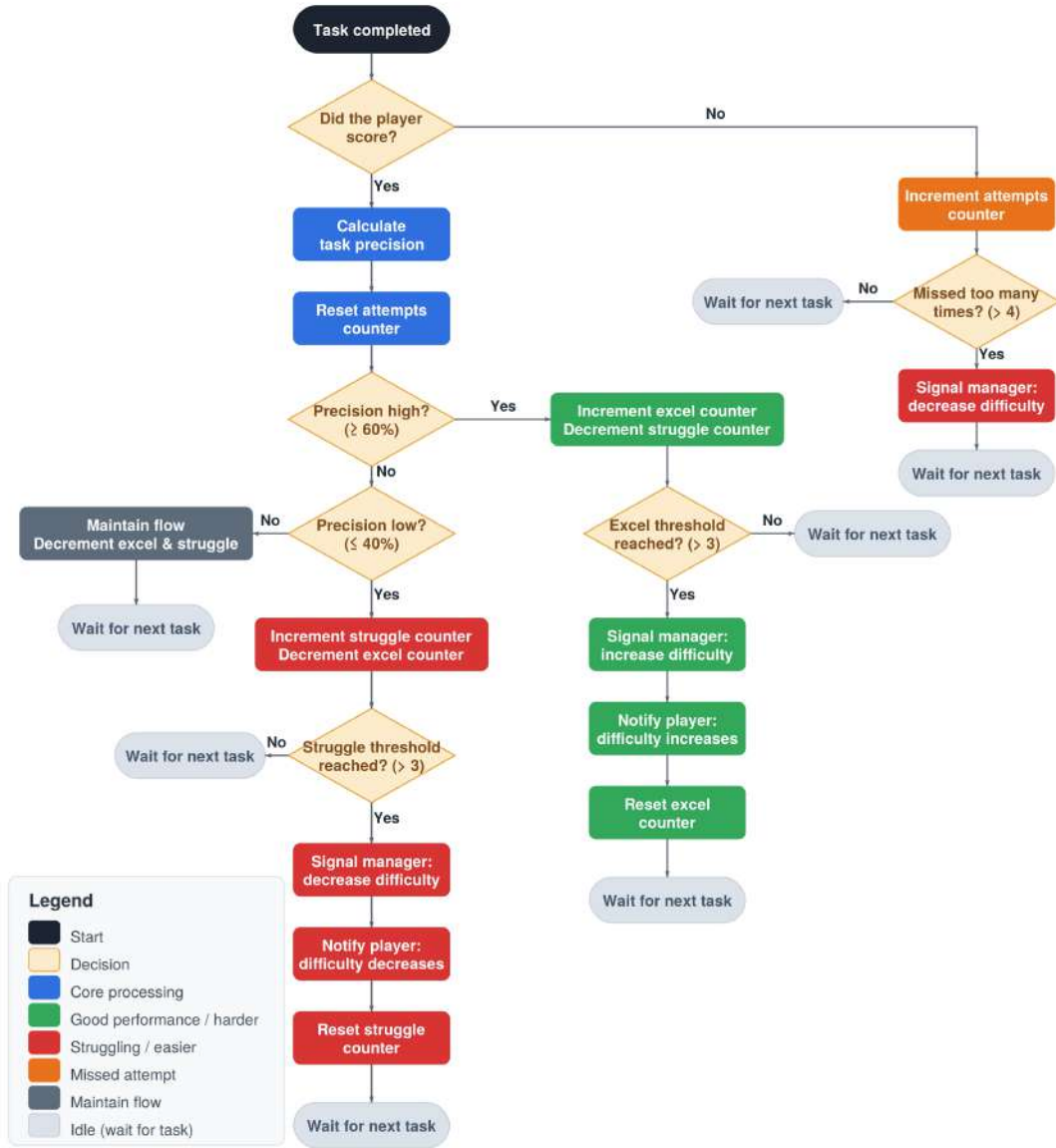


Figure 3.8: Flowchart of the *CarnyWise* DDA algorithm, illustrating the two main adaptation branches: the Failure Handling branch, which triggers a difficulty decrease upon four or more consecutive misses, and the Precision Analysis branch, which accumulates Excel and Struggle counters based on score precision thresholds ($\geq 60\%$ and $\leq 40\%$, respectively) to drive automatic difficulty adjustment.

Multimodal Difficulty Feedback

Providing immediate feedback is a crucial precondition for keeping players immersed in a state of flow. To reinforce this personalization and effectively communicate system state changes, the game utilizes a context-aware dialogue system driven by external **JSON** files. This system delivers dynamic and randomized motivational text feedback based on the user’s performance. Rather than explicitly stating that the difficulty has changed, the system provides subtle hints using common Portuguese colloquial expressions. For example, phrases such as “*Isso foi canja. Vamos passar às coisas sérias*” or “*Vamos aliviar um bocado. O*

importante é não parar”, are used to indicate an increase or decrease of the challenge level. It is worth noting that this feedback mechanism was not formally validated in terms of its direct impact on participants’ motivation levels. Rather, its purpose was to give players a clear sense that the difficulty was being adjusted, preventing them from unknowingly continuing to play based on an outdated difficulty level and, consequently, feeling frustrated by a mismatch between their performance and the difficulty they believed they were facing [8].

Furthermore, to complement the text displayed in a fixed location anchored to the player’s field of view, the system incorporates an **auditory cue** that is triggered whenever the difficulty level is adjusted. This sound provides players with immediate auditory feedback in addition to the visual indication, reinforcing their awareness that a change in difficulty has occurred, even if their attention is not directly focused on the displayed text at that moment.

Adaptation Mechanism

While *CarnyWise* dictates the timing of the adaptation, the specific adjustments are executed by the individual mini-game managers. To provide the infinite progression essential for long-term rehabilitation, the adaptation parameters are divided into two distinct categories as shown in Table 3.2:

- **Physical/Quantitative Parameters:** Parameters such as target distance and spawn quantity scale according to a code-based linear progression model ($y = mx + b$). This model provides practically unbounded difficulty scaling, subject to physical caps applied where clinically or spatially appropriate (e.g., the dog’s distance is capped at 14m and balloon density is bounded by the spawner area), ensuring the challenge remains progressive without becoming physically unreachable;
- **Cognitive/Complexity Parameters:** Parameters such as the probability of a target moving are managed using **Unity’s Animation Curve** feature (Figure 3.9). Despite the name, these curves are highly versatile and can be repurposed as difficulty curves, providing a smoother experience by continuously sampling difficulty values as the player progresses and supporting practically infinite progression, which is particularly suitable for long-term rehabilitation scenarios. Unlike the previous parameters, which rely on explicit mathematical formulas, the Animation Curve is defined in the **Unity Inspector** and used at runtime as a static, continuous difficulty function, conceptually similar to difficulty curves described in the literature. This visual authoring process allows developers and clinicians to specify non-linear difficulty progressions without requiring advanced mathematical or programming knowledge. Each parameter is governed by its own dedicated curve, allowing independent fine-tuning of its progression rate [40] [41].

Table 3.2: Adaptation Parameters per Minigame

Parameter Type	Frisbee Game	Archery Game
Physical/Quantitative Parameters	Target Quantity: <i>Infinite Scaling</i>: Increases by roughly 1 target every 5 levels. Distance: Linearly increases the dog's distance (capped at 14m).	Target Quantity: <i>Infinite Scaling</i>: Increases by roughly 1 balloon every 2 levels. Density: Increases the number of balloons that the balloon spawner can spawn.
Cognitive/Complexity Parameters	Movement: Percentage of targets moving laterally. Visibility: Percentage of targets blinking/disappearing.	Movement: Percentage of balloons moving. Transparency: Percentage of balloons becoming translucent (ghosts). Color Rule: The specific balloon color required to score changes dynamically with difficulty, preventing repetition. The HUD renders the required color rule text in the corresponding color to reinforce the visual association.

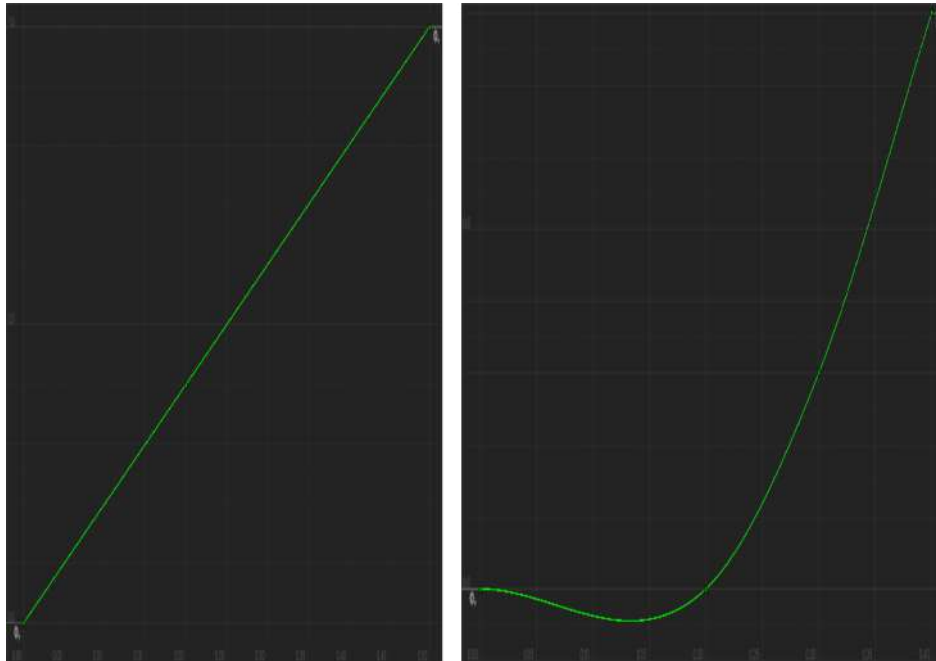


Figure 3.9: Example of two **Unity Animation Curves** used in the Archery Mini Game: the left one is used to configure the balloons that move, and the right one to configure the balloons whose transparency changes.

3.4 MINI-GAMES

The selection of the mini-games for the “**Country Fair**” was driven by a literature review on the therapeutic benefits of each activity, aimed at creating a comprehensive rehabilitation protocol. Rather than offering random activities, these specific mechanics were chosen to provide a complementary balance between static fine motor control and dynamic gross motor coordination, grounded in recent literature on adapted physical activities and specific requirements gathered from healthcare professionals. The two mini-games presented in this dissertation are part of a broader set of activities proposed, developed and refined in collaboration with healthcare professionals, whose clinical input was fundamental in defining, discussing, and improving the proposed concepts [39].

This section details how each mini-game serves as a complementary exercise for both upper limb motor function and cognitive rehabilitation, while also addressing other specific needs identified by healthcare professionals. It further describes the interaction mechanics and gestures required to play each game.

3.4.1 Common Interactions

Throughout the game, button-based interactions are performed using a **poke gesture**, in which the player extends their index finger as shown in Figure 3.10 to physically press a virtual button. This gesture was chosen due to its widespread adoption in **VR** applications as a natural and intuitive interaction for pressing buttons, and because it is natively supported by the **Meta Interaction Software Development Kit (SDK)** through its direct interaction system, reducing implementation complexity while ensuring a familiar user experience. This gesture is consistently used across all parts of the game, both for advancing dialogues and for navigating between areas, such as returning to the fair from a mini-game or selecting a mini-game from the fair.

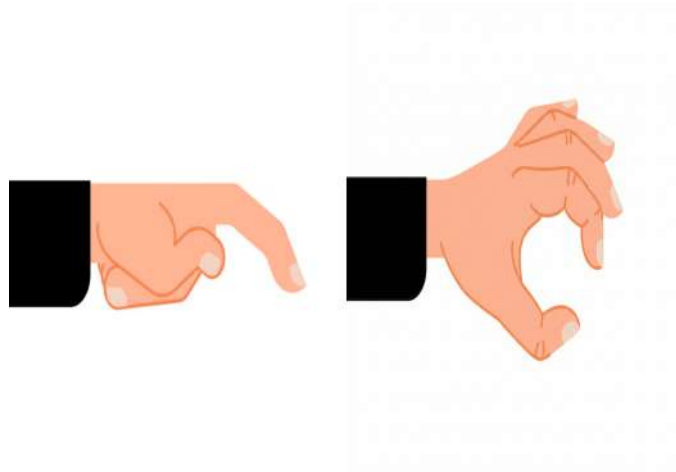


Figure 3.10: Interaction gestures used to interact with the game’s virtual environment: Poke gesture (left) Grab gesture (right). Created by Gemini Pro.

3.4.2 Archery Mini-Game

Upper Limb Motor Functions Justification

The act of drawing and holding a bow constitutes a highly effective bimanual static exercise that targets specific muscle groups differently in each arm. Recent electromyographic analyses demonstrate that archery heavily activates the biceps brachii and latissimus dorsi in the drawing arm, while the non-dominant bow arm relies significantly on the triceps brachii for elbow extension and the lateral deltoid for shoulder abduction. By replicating this precise muscular tension in VR through the “**grab**” gesture, the prototype aims to stimulate this complex neuromuscular engagement. The clinical benefits of this engagement are well-documented: research by Mahlovany et al. (2023) on adapted archery for Paralympic athletes demonstrated that specific archery training led to a reliable increase in handgrip force (rising from 37.4 kg to 45.5 kg) and improvements in the muscular endurance of the shoulder girdle. Furthermore, archery interventions have been proven to significantly improve hand flexibility, finger dexterity, and overall upper extremity motor functions in patients with neurological disorders, making it a highly suitable complementary physiotherapy for motor recovery [42] [43] [44] [45].

Cognitive Justification

Beyond the physical aspect, the archery mechanic inherently supports cognitive and psychological recovery. Successful archery requires a high capacity for **decision-making** and “afferent synthesis” within micro-intervals of time, contributing significantly to the athlete’s **psycho-emotional stability**. In fact, participating in archery exercises has been shown to alleviate depressive symptoms and improve overall psychological well-being by fostering a sense of self-fulfillment through hitting targets. Furthermore, recent neurorehabilitation research underscores a strong physical-cognitive interplay, noting that improvements in physical markers such as the handgrip strength developed during archery are positively correlated with enhanced cognitive performance, including memory and processing speed [43] [44] [45].

To further amplify this cognitive training, our prototype implements a dual-task paradigm by requiring users to pop only balloons of a specific color. This design choice is supported by the work of Bouatrous et al. (2023), who utilized color-matching tasks (placing fruits in corresponding baskets) to measure and train **selective attention**. Consequently, the **VR Archery** game serves as a composite exercise: the act of aiming trains emotional stability and decision-making, the physical grip strengthens critical neuromuscular pathways, and the color rule trains inhibition and selective attention [20].

Interaction Mechanics

In this mini-game, the bow is currently attached to the player’s left arm. The player uses their right hand to perform a grab gesture (Figure 3.10) to pull the bow’s string, as illustrated in Figure 3.11. The more they pull the string, the stronger the arrow’s launch force will be, although the bow can only be pulled up to a certain length. To launch the arrow, the player must open all fingers of their hand, releasing the string. An issue was observed regarding

hand-tracking during this gesture. Because the drawing motion pulls the string backward, when the hand moved behind the headset’s tracking sensors the string-pull was occasionally not detected, as the hand left the cameras’ field of view. During this mini-game’s tutorial, performing the gesture behind the sensors could, in some cases, lead to a crash of the game.

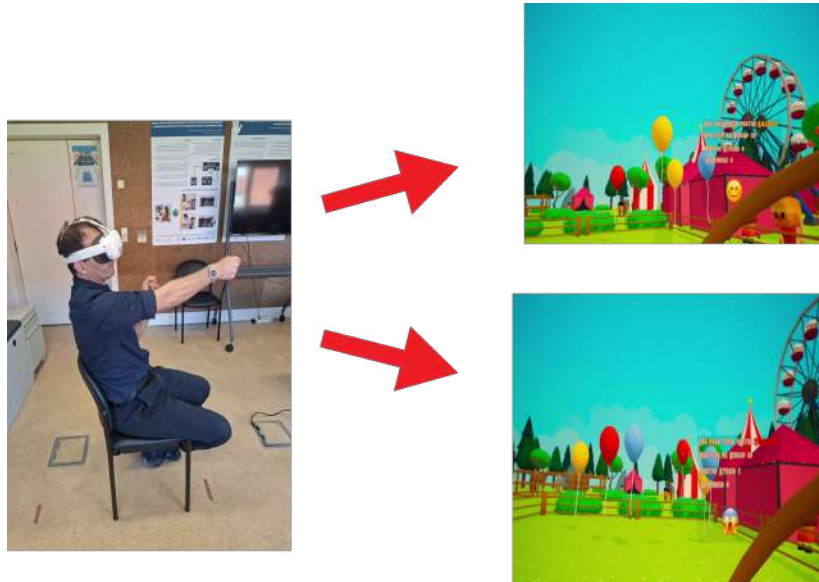


Figure 3.11: A participant performing the Archery mini-game gesture while seated, using the **Meta Quest 3** headset with hand-tracking interaction.

Figure 3.12 shows the forces acting on the arrow’s movement (launch, lift, gravity, and drag). These forces cause the arrow to follow a parabolic trajectory. When the arrow collides with any object, it automatically returns to the player’s bow, and the player must pull the string again to launch it. As mentioned before, to score points the player must pop a specific balloon color, which is displayed in the mini-game’s **HUD** alongside their score, streak, session goal, and the emotion representation provided by the **Emotion Recognition Module**, anchored to the player’s field of view to remain consistently visible regardless of head orientation. Additionally, there is an audience that surrounds the player and when they score, the audience performs a jumping animation followed by a spin in the air, providing positive visual feedback to encourage and keep the player motivated. Table 3.3a shows how the score varies depending on the balloon’s parameters.

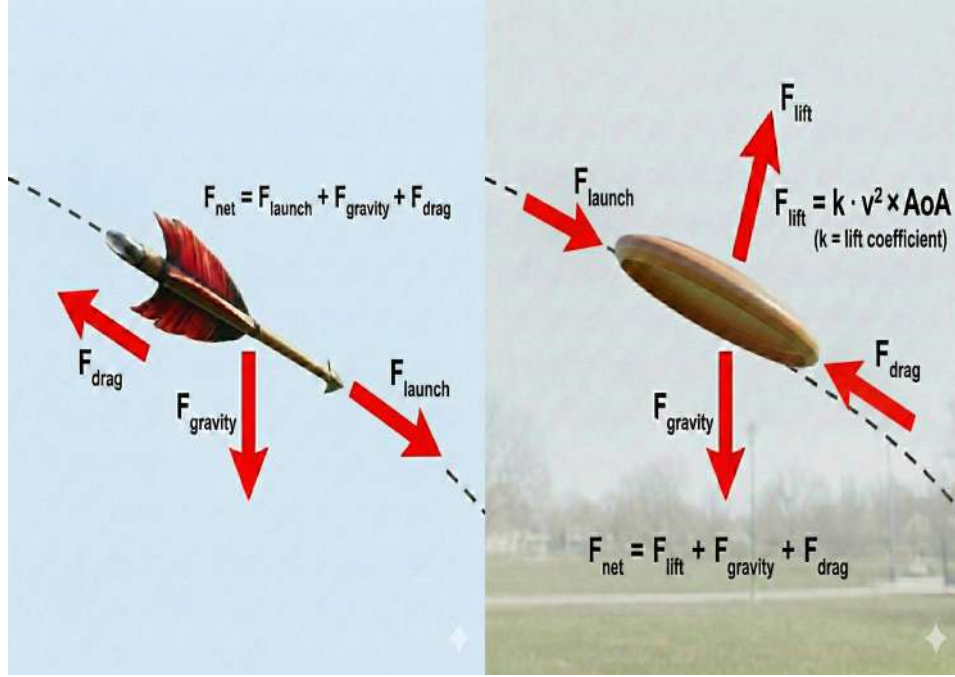


Figure 3.12: Schematic representation of the forces acting on the mini-game props during their motion: on the left, the arrow (archery mini-game) subject to launch force, gravity, and drag; on the right, the frisbee (frisbee mini-game) subject to launch force, gravity, drag, and lift. Created by Gemini Pro.

Table 3.3: Mini Games Score System

(a) Archery Mini Game

Balloons Parameters	Score Points
Static	1
Moves	2
Transparency Can Change	3
Moves And Transparency Can Change	6

(b) Frisbee Mini Game

Dog Area		Score Points
--		8

Other Score Areas Parameters	Score Points
Static	1
Moves	2
Can blink/Disappear	3
Moves And Can blink/Disappear	6

3.4.3 Frisbee Mini-Game

Upper Limb Motor Functions Justification

Although the user remains seated, the throwing motion requires a dynamic extension of the elbow and a coordinated flexion and extension of the wrist. This is clinically relevant because it combats the flexor synergy pattern (the tendency for the paretic arm to remain bent against the body), encouraging the user to extend the limb outwards.

Supported by the findings of Kao and Wang (2018), participating in Frisbee-based exercises requires significant upper limb engagement, specifically improving the functional abilities

of reaching, holding, and grasping. Furthermore, this specific training effectively enhances **upper limb power** which is defined as the product of muscle strength and speed as well as gross and fine hand motor dexterity. In the virtual environment, the survivor must not only hold the disc but release it at the precise moment of the swing to achieve the desired trajectory. This mechanic explicitly trains the timing of motor execution, directly rehabilitating these crucial upper extremity functions [46] [47].

Cognitive Functions Justification

Beyond motor skills, this mini-game addresses specific cognitive domains. Research by Kao and Wang (2018) highlights that Frisbee activities significantly improve **hand-eye coordination**, direction control coordination, and **reaction capability**. Moreover, a study by **Koutsobina et al. (2021)** emphasizes a strong correlation between perceptual-motor development (such as visual-motor integration) and the overall level of cognitive and mental functioning [46] [47].

In my prototype, the survivor must continuously calculate the distance to the dog or the secondary target zones, adjusting the force and angle of the throw accordingly. This process requires the brain to integrate visuospatial information with motor output in real-time. Furthermore, the requirement to react to the dog’s position and to the toggling visibility of the target zones trains processing speed and visual attention. This design effectively transforms the physical throwing gesture into a robust neurocognitive exercise that fosters both motor organization and cognitive development [46] [47].

The Role of the Dog

The inclusion of a virtual dog as the receiver is a design response to the specific needs identified in our survey with healthcare professionals [4].

The results indicated two critical requirements:

- **Guidance when Lost:** Professionals reported that **60%** of survivors often feel confused or lost in **VR**. To mitigate this, the dog acts as an active navigational cue. By barking or looking towards the target area, the dog directs the user’s attention, preventing disorientation without the need for intrusive text instructions [48] [49];
- **Social Presence:** The survey also emphasized the need to combat social isolation. Professionals suggested the use of “**animal-shaped NPCs**” to provide emotional support. The dog serves as a social agent, reducing the feeling of loneliness in the virtual world and creating an emotional bond that encourages the survivor to continue the therapy [4].

Interaction Mechanics

Upon entering the mini-game, the dog starts walking and chooses a random position in front of the player. The frisbee is positioned in front of the player’s head and follows its movement. This allows the player to grab the frisbee with their main hand by performing the gesture shown in Figure 3.10.

To throw the frisbee, the player must extend their arm and open all fingers of the hand they used to grab it, as illustrated in Figure 3.13. The quicker they perform these actions, the more intense the launch force will be, resulting in greater velocity. However, to make a more accurate throw, before extending their arm the player must bring the hand that grabbed the frisbee nearly to their opposite waist. In contrast to the archery gesture, no detection issues were observed when the hand was brought near the hip prior to the throw: the pinch pose was correctly identified in this position, and hand-tracking remained reliable throughout the throwing motion.

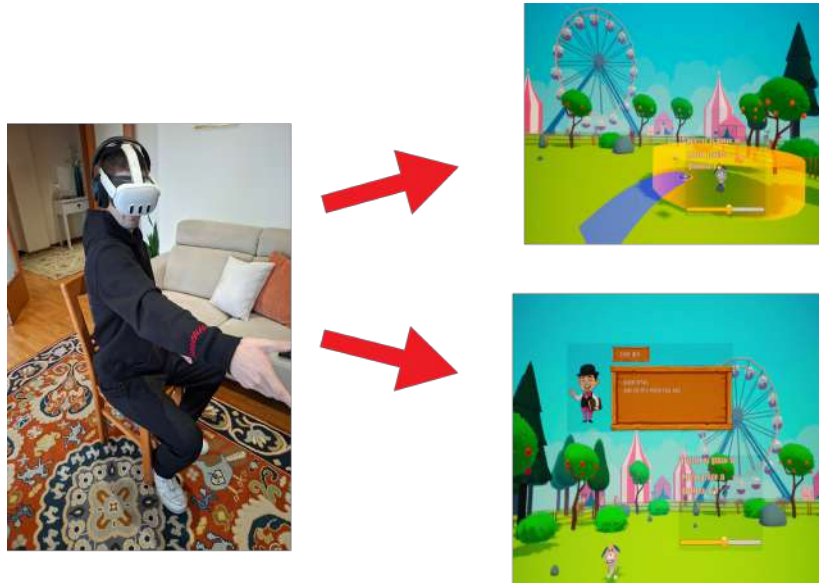


Figure 3.13: A participant performing the Frisbee mini-game throwing gesture while seated, using the **Meta Quest 3** headset with hand-tracking interaction.

Figure 3.12 shows the forces acting on the frisbee’s movement. These forces cause the frisbee to have a more linear trajectory at the beginning as a result of the frisbee’s lift force and then follow a more parabolic trajectory as it nears the ground.

As mentioned before, for the player to score points, the frisbee must land in an orange area surrounding the dog or in other orange areas. Table 3.3b shows the score values for each area and their parameters, which will be discussed in subsection 3.3.1. The player’s score, streak, session goal, and the emotion representation provided by the **Emotion Recognition Module** are displayed on the same **HUD** as the **Archery mini-game**, without the color rule indicator, anchored to the player’s field of view to remain consistently visible regardless of head orientation.

After the frisbee lands in one of these areas, all areas disappear, a sound score effect is played, and balloons appear, rise, and pop to encourage the player and keep them motivated. Following that, the dog jumps giving feedback to the player that it noticed the frisbee has landed and goes to its position to catch it. After catching it, the frisbee appears in the dog’s mouth, and delivers it to the player.

Once delivered, the dog chooses a random position in front of the player, and new target areas appear for the player to continue the game. However, if the frisbee does not land in one

of these scoring areas, the frisbee automatically reappears in front of the player.

3.5 EMOTION RECOGNITION MODULE

This module was not a primary objective of my dissertation, rather, it served as a complementary integration to verify whether the practical component of a colleague’s master’s dissertation (from the same research group) could be successfully merged with my prototype to create a more robust future application. My colleague’s (Bianca Guerreiro) research, titled “*Exploring Emotion Recognition for Stroke Rehabilitation: Towards Personalized Virtual Reality Serious Game*” aims to understand how emotion recognition can be utilized in personalized **VR** environments for post-stroke rehabilitation, exploring how detecting a user’s emotional state can foster a more adaptive and engaging therapeutic experience [38].

This collaboration had two main benefits. First, it served as a proof of concept that shared emotional awareness can be relayed into the rehabilitation environment, extending the system beyond motor and cognitive training toward the mitigation of social isolation. Second, and from an architectural standpoint, integrating an externally developed module validated the modularity of **Country Fair VR**: by consuming emotional state as a representation-agnostic output, the system accommodated a colleague’s independent research without modifications to its core, demonstrating extensibility toward a more robust future application.

The proposed architecture, illustrated in Figure 3.3, describes a scenario in which a stroke survivor’s emotional state is captured, transmitted to a remote server, and relayed to another survivor’s mini-game environment. Depending on the available specificity, the system communicates this state through one of three representation formats, selected randomly when entering a mini-game and adjustable via cheat codes(see Table 3.1): an *emoji-based representation*, which displays the recognized emotion as a simplified symbolic cue widely used as a substitute for emotional expression and shown to support rapid, intuitive interpretation; a *slider-based representation*, which reflects the emotion category (positive, neutral, or negative) as a position along a continuous positive-versus-negative scale initialized at a neutral value positive emotions shift the indicator toward the right, negative emotions shift it toward the left, and neutral emotions leave it unchanged as illustrated in Figure 3.14 and a *facial-expression-based representation*, which displays a reference image corresponding to the detected facial expression within the game environment, offering a more concrete and recognizable emotional cue. Table 3.4 lists all supported emotion types alongside their corresponding emojis, facial representations, and categories [38] [50] [51] [52].

Although the original concept envisioned capturing emotions directly through the **HMD**’s internal sensors, hardware constraints, specifically, the absence of a headset equipped with internal facial-tracking cameras, made this unfeasible. To replicate and validate the intended behavior, a laptop webcam was used instead to capture participants’ facial expressions and infer their emotional states, as shown in Figure 3.15.



Figure 3.14: Evolution of the emotion slider across three representative states: negative (left), neutral (center), and positive (right). Positive emotions detected shift the slider upward, negative emotions shift it downward, and neutral emotions leave it unchanged. The slider is initialized at a neutral value.

Table 3.4: All supported emotion types with their corresponding emojis, facial expression representations, and valence categories (Positive, Negative, or Neutral).

Emotion	Emoji	Face	Category
Happy	😊		Positive
Sad	😢		Negative
Angry	😡		Negative
Fear	😱		Negative
Surprise	😮		Positive
Disgust	😬		Negative
Neutral	😐		Neutral

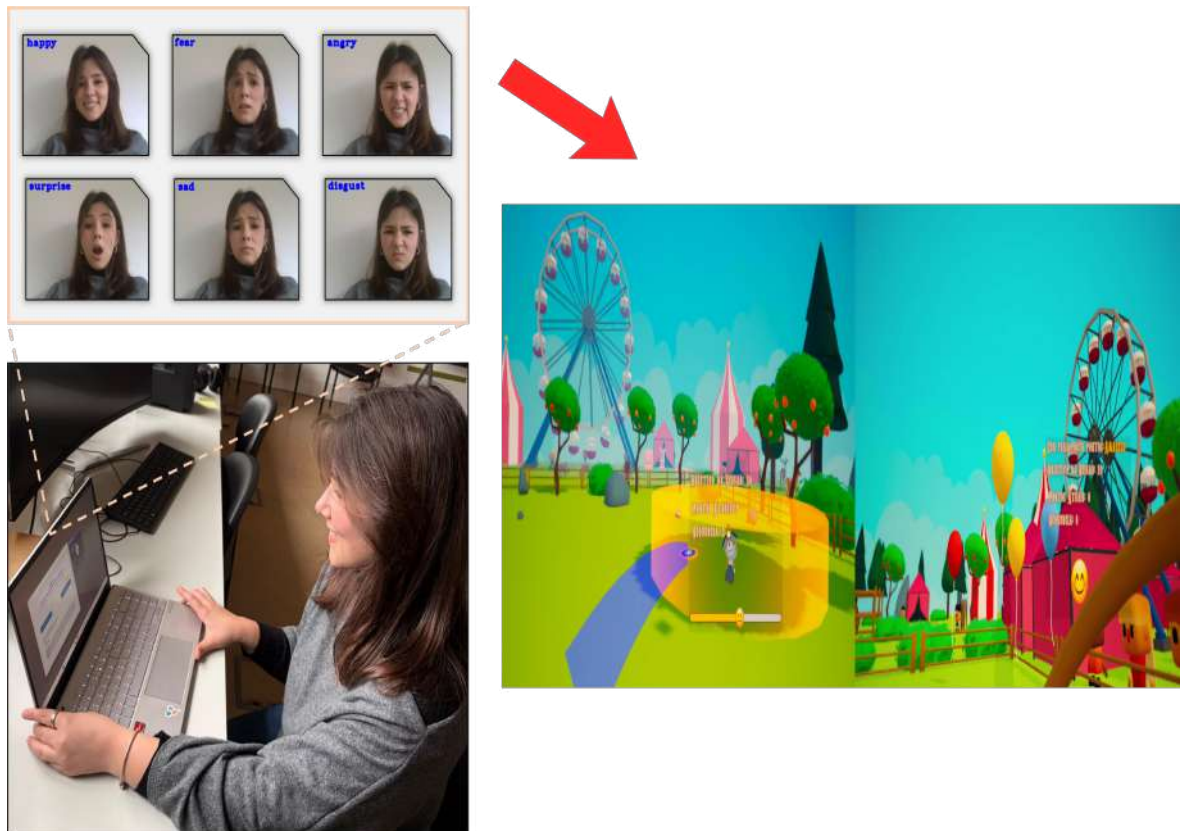


Figure 3.15: Illustration of the emotion recognition module: a laptop webcam captures the user's facial expressions (bottom-left), which are classified into one of the supported emotion categories (top-left), and the resulting emotion is relayed to the stroke survivor's VR mini-game environment as an emoji overlay, exemplified here by a happy emotion in the Frisbee mini-game and a neutral emotion in the Archery mini-game (right).

3.6 TECHNOLOGIES AND IMPLEMENTATION

This section details the technological foundation of **Country Fair VR**, describing the hardware platform on which the prototype runs, the software tools and engine used to build it, and the visual art direction that guided the production of its assets.

3.6.1 Hardware

At the hardware level, the prototype utilizes the **Meta Quest 3 Extended Reality (XR)** headset. This headset can recognize and interpret hand movements and gestures without the need for physical controllers, an advantage for individuals with neurological limitations or mobility issues affecting fine motor control. Furthermore, its robust tracking capabilities support the assessment and improvement of gross motor functions making it highly suitable for accurately detecting the wide-amplitude gestures required by the selected mini-games [53].

The hand-tracking latency on the **Quest 3** is reported at around **70 ms**, which the manufacturer classifies as *real-time* tracking². In practice, however, this classification does not guarantee an imperceptible delay: several users still report noticing a slight lag between physical movements and their reflection in the virtual environment. This residual latency, combined with the passthrough feed itself, can still contribute to cyber-sickness, meaning that low latency alone does not fully eliminate the risk of discomfort in **XR** applications.

Additionally, the hardware setup leverages 3D spatial audio to naturally guide the players through the virtual environment. While the use of external headphones is highly recommended to achieve maximum immersion, it is not mandatory, as the **Meta Quest 3** headset already features built-in speakers with robust spatial audio quality. Finally, the hardware configuration supports the connection of a peripheral keyboard (e.g., via Bluetooth) directly to the **HMD**. This addition is particularly useful for healthcare professionals, as it allows them to use cheatcodes (see Table 3.1) to manually manipulate the game state during runtime, such as adjusting the mini-game difficulty to suit a patient's needs³.

3.6.2 Software

The software was developed using the **Unity** game engine, which was chosen to facilitate and accelerate the development process through the **Meta XR Interaction SDK** a free, modular library for **Unity** and **Unreal Engine** designed to help developers create high-quality, intuitive, and natural hand and controller interactions in **VR**. **C#** was used for scripting the game logic, physics simulations, and the **DDA** algorithm.

Regarding the visual assets, while some 3D models and animations were custom-built using **Blender**, more detail about the 3D models will be discussed in Section 3.6.3. Furthermore, **FMOD Studio** was integrated to author and manage the project's audio. It was used both as the 3D spatializer for the spatial audio cues and to build multi-instrument/randomized

²Matts Digital. *Is Hand Tracking on the Meta Quest 3 Accurate and Responsive?* 2026. Available at: <https://knowledge.matts-digital.com/en/virtual-reality/meta/meta-quest-3/is-hand-tracking-on-the-meta-quest-3-accurate-and-responsive/> [Accessed: 27 March 2026].

³Steighner, M. *Meta Quest 3 Review I'm Finally Not Disappointed*. COGconnected, 2023. Available at: <https://cogconnected.com/review/meta-quest-3-review/> [Accessed: 27 March 2026].

containers that vary sound pitch and volume at playback, preventing auditory repetition and enhancing the overall immersive experience.

3.6.3 Visual Art Direction and References

To establish a coherent visual identity for the “Country Fair” environment, an art direction process was conducted prior to asset production. Reference imagery was gathered using the keywords *Carnival* and *Country Fair* on **Pinterest**, with the goal of identifying a suitable color palette, environmental props, and an overall aesthetic register consistent with the thematic setting. The collected references, illustrated in Figure 3.16, informed several key design decisions: the environment should adopt a vibrant, warm color palette, iconic fairground elements such as a Ferris wheel, red-and-white striped carnival booths, popcorn carts, balloons, and marquee tents should be present to reinforce environmental legibility and cultural familiarity.



Figure 3.16: Art direction reference board for the “Country Fair” environment. Environmental references (carnival tents, Ferris wheel, fairground props, and color palette) were gathered from Pinterest using the keywords *Carnival* and *Country Fair*. Additionally, the cover art and host character from the *Carnival Games* series (2K Play, Nintendo) informed the stylized, cartoon-like design of the game’s NPCs, while the disc dog activity from *Wii Sports Resort* (Nintendo, 2009) served as the primary gameplay and visual reference for the Frisbee mini-game.

Regarding the visual style of the non-player characters, **Zeca Bigodes** and **Carny Wise** were designed to evoke a style reminiscent of the host character from the *Carnival Games* video game series, favoring a stylized, cartoon-like aesthetic suited to the rehabilitation context. Both characters, along with their respective dialogue box layouts, were generated using the image creation feature provided by **Gemini Pro** (Google).

The **Frisbee mini-game** drew direct inspiration from the disc dog activity featured in *Wii Sports Resort* (Nintendo, 2009), in which the player throws a frisbee towards designated areas for a dog to retrieve. This reference also informed the initial low-poly, rounded art style

adopted for that mini-game’s assets, which was subsequently generalized to the entire project to ensure visual consistency across all environments and interactive elements.

The majority of the 3D models used in the environment were sourced from Poly Pizza, an online repository of free, low-poly assets. Table 3.5 lists each model, its author, and the corresponding attribution link.

A set of 3D models was contributed by **Ana Rita Guimarães**, a colleague from the *Virtual and Augmented Reality* course at **Universidade de Aveiro**, whose work was developed as part of the same prototype used in the first wave of user testing (Section 4.1). Her contributions include the 3D models of the bow, arrows, and balloons, the core interaction mechanics of the Archery mini-game specifically, drawing and releasing the bowstring to fire arrows and the balloon burst behavior upon arrow impact as well as the Ferris wheel rotation animation and its cabin attachments.

Table 3.5: Assets sourced from Poly Pizza, with respective authors and attribution links.

Asset	Author	Link
Sheep Model	Google	poly.pizza/m/dXBMV4AY2DL
Ferris Wheel Model	Google	poly.pizza/m/5KiVEnXN5Cw
Tent Model	Google	poly.pizza/m/9Ob6OO8HMjX
Fair Animals Models and Animations (Bundle)	Quaternius	poly.pizza/bundle/Animated-Animal-Pack-ILAPXeUYiS
Persons Models and Animations (Bundle)	J-Toastie	poly.pizza/bundle/CUTES-Part-One-WD91WrT0gx
Carnival Booth Model	sirkitree	poly.pizza/m/4NizXJZsuO2
Merry Go-Round Model	sirkitree	poly.pizza/m/8BliGG-e55g
Ground Model	Adam Tomkins	poly.pizza/m/achm-Cr9Rr3
Big Tent Model	Ian MacGillivray	poly.pizza/m/0TmXixFN81A
Popcorn Cart Model	Don Carson	poly.pizza/m/1T9untmohmj
Bench Model	Sammy	poly.pizza/m/3ytpP2GLb0b
SkyBox Texture	Borodar	assetstore.unity.com/.../farland-skies-cloudy-crown-60004

The 3D button models used across the interface were sourced from the **Meta XR SDK**. The remaining assets the frisbee, the dog, and the ambient environment props (trees, rocks, and bushes) were modelled and rigged by the author (Figure 3.17). The dog model additionally received custom *Idle* and *Walk* animations, also authored from scratch using **Blender’s** armature system.



Figure 3.17: 3D assets modelled and rigged by the author in **Blender**: the **NPCs**, comprising the original character model sourced from Poly Pizza (left) and its modified version retextured to adopt a carnival worker aesthetic (center-top), the frisbee model (right-top); the ambient environment props, comprising trees, bushes, and rocks (bottom-left) and the dog model with its full rig, used to drive the custom *Idle* and *Walk* animations developed for the Frisbee mini-game (bottom-right).

User Study

This chapter describes the assessment moments of the proposed personalized **Virtual Reality (VR)** based country fair, reports the main insights collected, and discusses the results in light of the topic being addressed.

4.1 FIRST WAVE: INFORMAL USABILITY TESTING

During the early development phase, informal usability testing was conducted as part of a **virtual and augmented reality** course at **University of Aveiro**, during which students and teachers had the opportunity to interact with the prototype and provide feedback. Across the demonstration sessions, approximately six to eight individuals engaged with the system. The primary objectives of this first wave of testing were threefold: to evaluate whether the prototype successfully conveyed the intended vibrant “**Country Fair**” environment, to evaluate overall enjoyment and engagement.

4.1.1 Prototype State

At the time of this evaluation, the prototype was in an early functional stage as shown in Figure 4.1. The following describes its state, providing context for the scope and limitations of the test:

- **Game Environment:** The game environment was already populated with **Non-Playable Characters (NPCs)** and thematic fair elements (e.g., tents, ferris wheel and balloons), establishing the intended aesthetic;
- **Fair Scene:** This scene was already implemented with the tents to choose the mini-games;
- **Mini-Games:** Two mini-games (archery and frisbee) were implemented;
- **Game Objectives:** Neither mini-game had a defined end state;
- **Scoring:** In the balloon archery mini-game, balloons of different colors carried different point values (yellow: 3 points, red: 2 points, blue: 1 point);

- **HUD:** The in-game **Heads-Up Display (HUD)** was rendered at a fixed world-space position, rather than being anchored to the player’s field of view;
- **Personalization Module:** Not yet implemented, and neither had adaptive parameters been defined nor integrated;
- **Cheatcodes:** To accelerate iterative testing without requiring a full playthrough, a set of keyboard shortcuts was introduced at this stage. With the exception of the emotion representation commands (`positive/negative` and `f(expression_name)`), all cheatcodes listed in Table 3.1 were implemented at this point;
- **Emotion Recognition Module:** Not yet implemented.



Figure 4.1: Screenshots of the prototype evaluated during the first wave of testing: the Country Fair scene (top), the Archery mini-game (bottom left), and the Frisbee mini-game (bottom right).

4.1.2 Experimental Setup

Participants interacted with the game from a seated position using the **Meta Quest 3 Head-Mounted Display (HMD)**, the primary hardware defined during the development phase. To allow for clearer perception of the game’s 3D spatial audio and to maximize immersion, users were equipped with **Sony WH-CH500** wireless headphones, as shown in Figure 4.2



Figure 4.2: First Wave informal usability session: two participants interacting with the **Country Fair VR** prototype from a seated position using the Meta Quest 3 HMD and Sony WH-CH500 wireless headphones.

4.1.3 Measurements

Prior to the sessions, a set of environmental checks was performed, including verifying lighting conditions and ensuring participants had sufficient space to move freely without risk of collision. Throughout the sessions, an observer monitored interactions and systematically recorded any usability issues or difficulties encountered.

4.1.4 Procedure

This evaluation was carried out at the **Virtual and Augmented Reality Research Group (VAR Lab)** at **Institute of Electronics and Informatics Engineering of Aveiro (IEETA)**, in the scope of the Virtual and Augmented Reality course, involving both students and teachers as participants. The assessment of the prototype followed a structured sequence of steps:

- **Setup:** Initial preparation of the room and configuration of the required equipment, including the HMD and headphones;
- **Briefing:** A brief introduction was provided, covering the project's context, the game's objectives, and the specific metrics to be evaluated during the session;
- **Gameplay:** Each participant interacted with the game prototype under observation;
- **Debriefing:** Upon completing the gameplay session, the observer invited participants to share their overall experience, suggestions, and feedback.

4.1.5 Results and Feedback

Participants were able to fulfill the proposed objectives and successfully use the system, exploring both mini-games without significant difficulty. Observations indicated that users were visibly engaged throughout the experience, interacting with the system with little to no assistance required. No signs adverse effects, such as cybersickness, were observed, confirming the safety of the interaction design.

Following the sessions, participants provided qualitative feedback that was categorized to guide iterative refinement of the prototype:

- **Overall Experience:** The mini-games were considered fun to play, and participants reported feeling immersed in the “**Country Fair**” environment. The auditory feedback was particularly appreciated; however, the absence of an in-game tutorial was strongly noted as a limitation;
- **VR Frisbee Feedback:** One participant noted that the frisbee curved excessively towards the throwing hand, and felt the dog’s retrieval mechanic was slightly unnatural, suggesting the dog should deliver the frisbee directly to the nearest outstretched hand;
- **VR Archery Feedback:** A physical glitch was reported in which the arrow would sometimes launch while the bowstring was still being drawn. Additionally, using textures or legend plates to distinguish scoring balloons rather than relying solely on color — was suggested.

Overall, this first wave provided an initial understanding of key aspects such as usability, interaction quality, and participant reaction to the Country Fair approach, informing the refinements carried out prior to the second wave of user testing.

4.2 SECOND WAVE: FORMAL USABILITY TESTING WITHOUT THE TARGET AUDIENCE

After developing a robust prototype with integrated personalization features, a formal usability test was conducted with **38** participants. The primary objectives of this testing phase were:

1. To determine whether participants noticed the automatic personalization system in action and if they felt the game’s difficulty adapted appropriately based on their performance;
2. To evaluate the user experience, usability, interaction and design of the system;
3. To verify other critical factors before proceeding with the target audience (stroke survivors), such as whether the mini-games’ instructions and goals were clear and easy to understand, and whether participants experienced physical fatigue or felt overwhelmed during gameplay;
4. To review some of the core metrics from the previous testing wave (i.e., whether participants felt they were at a country fair, and whether they felt motivated, engaged, and had fun while playing).

4.2.1 Prototype State

The prototype tested in this wave (Figure 4.3) incorporated a set of targeted refinements driven by the feedback gathered during the first wave. Addressing the participant’ feelings of being lost, a comprehensive tutorial and practice feature was implemented before each mini-game. The physical glitch identified in the archery mini-game, where the arrow would prematurely launch while the string was still being pulled, was debugged and successfully resolved. Furthermore, the feedback regarding the archery scoring system led to a significant therapeutic improvement: instead of adopting the complex suggestion of using textured

balloons, the system was redesigned to incorporate a new cognitive rule whereby the player must shoot a balloon matching a specifically requested color. This design choice not only resolved the scoring ambiguity but also elevated the mechanic into an engaging cognitive exercise targeting selective attention, suitable for stroke survivors [20].

Beyond these corrections, the following describes the state of the prototype at the time of this evaluation:

- **Game Environment:** The fair environment was already populated with **NPCs**, fair workers, and fair animals, alongside additional thematic props (benches, tents, etc.). A post-processing effect was also applied to the virtual environment, elevating contrast and saturation levels to reinforce the carnival aesthetic;
- **Player Hands:** For the **VR** representation of the player’s hands, the texture was replaced with a more realistic one instead of the default shadow hands provided by the **Meta Extended Reality (XR) Software Development Kit (SDK)**, in order to provide a more immersive experience for the participants;
- **Fair Scene:** The fair scene included the game introduction sequence delivered by **Zeca Bigodes**, the mini-game selection tents, and the motivational congratulatory messages displayed upon completing a mini-game, at which point the player is teleported back to the fair;
- **Game Objectives:** Both mini-games had a defined end condition: the player was required to reach a predefined target score to complete the session;
- **Scoring:** Both mini-games already featured their final scoring system, as defined in Table 3.3;
- **HUD:** The in-game **HUD** was still rendered at a fixed world-space position;
- **Personalization Module:** Fully implemented as described in Subsection 3.3.1, with the exception that the text feedback was rendered at a fixed world-space position rather than being anchored to the player’s field of view;
- **Emotion Recognition Module:** Implemented with the emoji representation form only, the facial expression and slider representations were not yet integrated. This module was not actively used during this testing wave.



Figure 4.3: Screenshots of the prototype evaluated during the second wave of testing: the Country Fair scene (top), the Archery mini-game (bottom left), and the Frisbee mini-game (bottom right).

4.2.2 Experimental Setup

The experimental setup was highly similar to the one used in the first wave, with the main differences being the audio device and the inclusion of a peripheral keyboard. While headphones were mandatory in the first test to verify the 3D spatial audio functionality, their use was optional in this phase. Furthermore, the **Sony WH-CH500** wireless headphones were replaced by a **Steel Nova 3 Wireless** headset, as this newer device provides superior 3D spatial audio quality.

Additionally, a **Bluetooth keyboard** was connected to the **HMD**. This keyboard was not intended for player interaction rather, it allowed the observer to input '**cheat codes**'. For example, these codes were used to skip tutorials for participants who had already watched others play and felt confident proceeding, or to manually force a change in the mini-games' difficulty. Manually triggering difficulty changes was crucial for this test, as the algorithm's default adaptation might perfectly match a user's performance, which would prevent them from experiencing and reporting on noticeable difficulty shifts.

4.2.3 Tasks

To address this test objectives, participants were asked to perform a sequence of three specific tasks within the VR environment. These tasks consisted of selecting a mini-game, completing the frisbee mini-game, and completing the archery mini-game.

4.2.4 Measurements

Before the gameplay sessions began, certain measurements were taken, such as checking the room’s lighting and ensuring that participants would not hit any surrounding objects while playing. During the sessions, participants were continuously observed by one to two observers, who recorded any difficulties encountered and verbal comments made. Afterward, participants completed a comprehensive questionnaire. The questionnaire began with a brief description of its objectives, assuring participants that all collected data would be used exclusively for academic purposes. It primarily consisted of mandatory multiple-choice questions with seven options using a Likert scale. These questions were organized into the following sections: *Demographic Data* (e.g., gender, age, prior experience with **VR**), *About Personalization*, *Engagement & Motivation*, *Comfort & Trust* and *Feedback & Understanding*.

4.2.5 Procedure

The majority of the user tests were carried out at the **VAR Lab** at **IEETA**, while some were conducted in private homes. The evaluation of the prototype followed a structured and logical process based on the following steps:

- **Setup:** Initial preparation of the room and configuration of the required equipment (HMD, headphones, and the Bluetooth keyboard);
- **Consent:** The participant agreed to take part in the experiment and provided informed consent, including permission to be photographed during gameplay;
- **Briefing:** A short introduction was provided regarding the project’s context, the game’s objectives, and the specific metrics being evaluated during the test;
- **Gameplay:** The participant tested the game prototype;
- **Debriefing:** Upon completing the gameplay session, the participant answered a post-experience questionnaire.

Additionally, the questionnaire included two optional sections: an *Acceptance* section, which asked whether participants felt the product was suitable for stroke rehabilitation, and *Effectiveness & Perceived Value* section where participants were asked to put themselves in the shoes of a stroke survivor and indicate whether they believed this personalized experience could support their rehabilitation progress. Both sections were made non-mandatory for the same reason, some participants without medical backgrounds might lack the necessary clinical knowledge to answer confidently.

Finally, a *Suggestions or Feedback* section provided three open-ended questions for participants to share qualitative suggestions, general impressions of the game, and impressions of the personalized **XR** experience.

As previously mentioned, assessing the system’s usability was a primary goal. To achieve this, the mandatory *System Usability Evaluation* section incorporated the **System Usability Scale (SUS)**, a standard measuring instrument consisting of 10 generic statements that enable a robust evaluation of the application’s usability, specifically focusing on aspects such as effectiveness, efficiency, and overall user satisfaction. Since each testing session already

comprised a gameplay period followed by the completion of the questionnaire, the SUS was the only standardized instrument administered, in order to keep the sessions from becoming too long for participants.

4.2.6 Participants

A total of 38 participants were recruited for the study, with ages ranging from 21 to 57 years (6 female, 32 male). Among them, 17 had prior experience with **VR**.

The majority of participants were students (32 out of 38), most of whom were studying computer science related fields, including two who held a research grant in **XR** development. Of the remaining six participants, one was unemployed, and the other five came from healthcare and technology backgrounds: two nurses, one personal trainer, and two professionals from the IT area specifically, an **XR** technology university professor and an **Artificial Intelligence (AI)** engineer. Their diverse expertise was valuable in ensuring well-rounded feedback.

Figure 4.4 shows 15 of these participants playing the game prototype.



Figure 4.4: Second Wave formal usability study: a representative sample of 15 of the 38 participants interacting with the **Country Fair** VR prototype from a seated position, evaluating system usability, perceptibility of the **Dynamic Difficulty Adjustment (DDA)** personalization system, and overall engagement with the “Country Fair” environment.

4.2.7 Data Analysis

To support the analysis of the collected data namely results, suggestions, feedback, and observations **Python** scripts were developed with the assistance of generative **AI** tools. These scripts use regular expression (**regex**) **pattern matching** to automatically classify free-text responses into predefined thematic categories, allowing the review process to be carried out more efficiently. The automatically generated outputs were subsequently subject to manual verification, in order to minimize categorization errors. The analysis of the quantitative data was primarily descriptive, focusing on response frequencies and the aggregated SUS score. No inferential or correlational analyses were performed, as the study was designed as a usability evaluation rather than a hypothesis-driven experiment.

4.2.8 Observations and Results

During the testing sessions, observers recorded notable behaviors and verbal comments expressed by participants, providing valuable qualitative insights that complement the quantitative questionnaire data.

The most notable observation was that **26 participants (68%)** explicitly expressed a desire to keep playing after the formal test had ended. This spontaneous willingness to continue engaging with the prototype strongly corroborates the questionnaire findings regarding motivation and engagement, suggesting that the rehabilitation game successfully avoids the monotony often associated with conventional therapeutic exercises.

Regarding the primary objective of evaluating the personalization system, the results were overwhelmingly positive. When asked whether they noticed the system adapting the mini-games during the session, **31 out of 38 participants (82%)** responded positively, while only 5 (13%) responded negatively and 2 remained neutral. Furthermore, regarding the statement “*The system adapted the mini-games to my specific abilities and needs*”, **33 participants (87%)** expressed agreement, with only 1 disagreement and 4 neutral responses. These findings confirm that the dynamic difficulty adjustment mechanism was functional and perceptually salient to the vast majority of users, fulfilling a core design requirement of the prototype. It should be noted, however, that these results capture the *stated perception* of adaptation rather than its measured impact on performance or experience, isolating the latter would require a controlled comparison against a non-adaptive condition.

Beyond personalization, the results also validated several other critical factors. Regarding interface clarity, **31 participants (82%)** agreed that “*The instructions and objectives of the mini-games were clear and easy to understand*”, with only 2 disagreements and 5 neutral responses, indicating that the newly implemented tutorial system effectively addressed the confusion reported during the first wave of testing. Physical comfort was confirmed as a non-issue: **37 participants (98%)** did not feel overly tired or overwhelmed during the session, with only one negative response a particularly important outcome given that prolonged sessions are often necessary in a rehabilitation context.

Environmental immersion and motivational quality also received strong endorsement. **34 participants (90%)** agreed that they felt immersed in a country fair environment, with

the remaining 4 remaining neutral and no disagreements recorded. Most importantly, **37 participants (98%)** responded positively to the statement *The country fair environment created a fun, motivating, and engaging experience*”, with only one neutral response. These results suggest that the Country Fair” aesthetic successfully promoted engagement and enjoyment, which are essential drivers for adherence in stroke rehabilitation. However, it is worth noting that immersion should be carefully managed in terms of session duration, as prolonged exposure to virtual reality may lead to fatigue or discomfort, particularly in populations with neurological impairments.

The prototype achieved a remarkable average **SUS score of 82 (out of 100)**. According to the evaluation framework proposed by Bangor et al. (2009), this score falls within the **“Excellent”** range (80.0–89.9), corresponding to a letter grade of **A- to A**, well above the “Good” threshold (70.0–79.9). In practical terms, this rating suggests that users found the **VR** rehabilitation game intuitive, easy to use, and well-suited for its intended purpose, with no significant barriers to interaction. Given that the SUS has been validated as a robust metric for evaluating immersive XR applications in healthcare, this outcome strongly reinforces the quality of the implemented design and its readiness for subsequent testing with the target audience [54] [55] [56].

Regarding secondary objectives, the spatial sound feedback item yielded mixed but generally positive results: **25 participants (65%)** responded positively regarding whether *the spatial sound feedback was helpful in guiding my actions*, while **11 (30%)** remained neutral and **2 (5%)** responded negatively. This result is important as it may help determine whether this feature could be beneficial in cases where survivors feel lost during gameplay [4].

The non-mandatory sections *Acceptance* and *Effectiveness & Perceived Value* yielded exclusively positive or neutral responses, with no disagreements recorded across any of their items. For the *Acceptance* section, **37 participants** responded. Regarding *I believe that personalized VR could be helpful for stroke rehabilitation*, **32 participants (86%)** responded positively and **5 (14%)** remained neutral. For *I believe this experience could serve as a beneficial complement to traditional rehabilitation programs by reducing monotony*, **36 participants (97%)** responded positively and **1 (3%)** remained neutral.

For the *Effectiveness Perceived Value* section, **27 participants** responded to *I believe that personalized XR could support my rehabilitation progress*, with **22 (81%)** responding positively and **5 (19%)** remaining neutral. For *The personalized experience felt beneficial for my rehabilitation*, **28 participants** responded, with **20 (71%)** responding positively and **8 (29%)** remaining neutral. It is important to note that these participants were not part of the target population neither stroke survivors nor healthcare professionals and therefore these results should not be interpreted as clinical evidence. Nevertheless, the consistently positive responses, with no negative ratings recorded, provide an encouraging indication of the system’s perceived value and support the relevance of further evaluation with the intended end users.

Regarding interaction mechanics, **13 participants (34%)** experienced difficulty with the frisbee mechanics, particularly in picking up, releasing, and throwing the disc. While

the underlying physics simulation was correctly implemented, the dissipative forces (e.g., air resistance) are handled automatically by **Unity’s** physics engine, and the initial throw velocity is calculated by the **Meta XR SDK** based on the user’s hand movement. Since the developer lacks full direct control over these components, their interaction can introduce variability perceived as instability particularly relevant in rehabilitation, where consistent and predictable mechanics are essential for building patient confidence. By contrast, **9 participants (24%)** reacted positively to the frisbee game, and **14 (38%)** to archery.

Furthermore, **4 participants (11%)** encountered difficulties with archery mechanics, primarily related to arrow nocking and string pulling. Notably, **9 participants (24%)** did not notice the adaptive color-based scoring rule or the color change mid-game, suggesting that the visual indicator within the **HUD** was not positioned optimally, or that the relationship between difficulty changes and balloon color changes was not sufficiently apparent. Technical issues such as bugs and crashes were observed in **7 participants (18%)**.

Suggestions and Feedback

From the **38** participants, **31 provided written suggestions or feedback** regarding their experience with the prototype. These qualitative responses were analyzed and categorized into key themes, offering valuable insights for future iterative refinements.

The most prevalent theme, mentioned by **21 participants (68%)**, was that the game was praised as being fun, engaging, and intuitive to play. Closely related, **9 participants (29%)** expressed a positive reaction to the immersive quality of the “Country Fair” environment, reinforcing the success of the game’s aesthetic design.

Regarding specific game mechanics, the most frequent point of criticism concerned the **frisbee mechanics**, which **11 participants (35%)** described as awkward, inconsistent, or physically unintuitive. Furthermore, **7 participants (23%)** reported that on-screen text or instructions were hard to read or too lengthy, and **3 participants (10%)** noted that the adaptive color-based scoring rule for archery was not obvious or easy to notice. These issues suggest that text elements were not positioned correctly or that the font was not properly adapted for **HMDs**, resulting in a slightly pixelated appearance. Proposed solutions included adding audio narration or voice-over for instructions, and placing the **HUD** and text boxes closer to the player. A further mechanical issue, reported by **5 participants (16%)**, concerned difficulty pressing in-game buttons, likely due to inconsistent spacing between them and imprecise capture of the poke gesture (Figure 3.10) for some users.

4 participants (13%) suggested the addition of more mini-games to enhance engagement, proposing examples commonly found at country fairs such as a duck picking game or a fishing game. Regarding personalization specifically, **2 participants (5%)** suggested adding more suggestive audio cues when difficulty changes (e.g., bell or level-up sounds), and another **2 (5%)** proposed that maintaining a scoring streak could grant bonus points. One participant recommended that the frisbee mini-game display a trajectory line similar to archery, which could disappear at higher difficulty levels, while another made the same suggestion for the archery game itself. Additionally, one participant suggested providing explicit feedback on difficulty progression at the end of the session, such as a high-score comparison or an in-game

indicator rewarding improved performance.

4.2.9 Limitations

Despite the encouraging results, this evaluation is not without limitations. Regarding interaction design, in the Archery mini-game the virtual bow is currently attached to the player’s left arm, with the right hand performing the grab gesture to pull the string. **This VR interaction is not optimal for a rehabilitation scenario, since stroke survivors may be paralyzed in their left arm.** This configuration was implemented for prototype purposes only. Despite this known limitation, the mini-game remains therapeutically validated by the upper limb motor and cognitive benefits discussed in Subsection 3.4.2, which are independent of the arm holding the bow.

More critically, both formal waves of testing were conducted exclusively with a **non-clinical population** healthy adults, predominantly students and technology professionals. Although the diversity of expertise provided well-rounded feedback, the absence of the true target population post-stroke survivors represents the most significant limitation of the present evaluation. The planned clinical validation with survivors at the **Rovisco Pais Rehabilitation Center**, the primary partner institution for this project, could not be carried out within the scope of this dissertation due to logistical and timeline constraints. Consequently, while the prototype has demonstrated strong usability and motivational properties with a general-population substitute, its therapeutic efficacy, accessibility for motor-impaired users, and clinical safety profile remain empirically unvalidated.

4.3 PARALLEL STUDY: EMOTION VISUALIZATION EVALUATION

The three emotional representation formats integrated into the *Emotion Recognition* module described in Section 3.5 were the subject of a dedicated comparative evaluation conducted in parallel by a colleague from the same research group (Bianca Guerreiro), carried out at the **VAR Lab** at **IEETA**, as part of her master’s dissertation. For this evaluation, the author contributed by integrating the *Personalized VR Country Fair* as a client of the emotion recognition pipeline, adapting the in-game user interface to support the three representation formats under comparison, and accompanying the testing sessions as an observer, ensuring the correct functioning of the integrated system throughout.

The study aimed to evaluate how the **emoji-based**, **slider-based**, and **facial-expression-based** representations were perceived by users in terms of usability, ease of interpretation, social-emotional connection, and perceived accuracy. A total of **17** participants rated all three formats within a within-subjects design, using a 7-point Likert scale across 14 items grouped into four analytical dimensions.

4.3.1 Prototype State

The prototype evaluated in this parallel study is the final version, as illustrated in Figure 3.5, incorporated a set of targeted refinements driven by the feedback and observations gathered during the second wave of testing. Addressing the reported difficulty in reading

on-screen text and noticing the adaptive color-based scoring rule, the **HUD** was redesigned to follow the player’s head movement, ensuring that all relevant information remained consistently within the field of view. Furthermore, to improve the visibility of the target color rule in the archery mini-game, the **HUD** text indicating the required balloon color is now rendered in its corresponding color for example, if the target color is blue, the text is displayed in blue reinforcing the visual association between the scoring rule and the balloon to hit. Additionally, a recurring edge-case bug in the archery mini-game, where the arrow after launch would leave the play area and fail to return to the bow, was identified and resolved.

Beyond these corrections, the following describes the state of the prototype at the time of this evaluation:

- **Personalization Module:** The sole change relative to the previous wave was anchoring the text feedback to the player’s field of view. This adjustment ensures that difficulty change notifications are always visible to the player, regardless of their head orientation, thereby reinforcing the perceptibility of the **DDA** system and reducing the risk of users missing adaptation events;
- **Cheatcodes:** Implementation of the cheatcodes:(`positive/negative` and `f(expression_name)`), see Table 3.1;
- **Emotion Recognition Module:** The slider and facial expression forms of emotion representation were implemented, complementing the emoji representation already present in the previous wave.

4.3.2 Results

Results indicated that all three formats were generally well received, with mean scores above the neutral midpoint across all dimensions. The **emoji-based** and **slider-based** representations performed comparably regarding usability and ease of interpretation, while the **facial-expression-based** format obtained notably higher scores in the social-emotional connection dimension, particularly in empathy and perceived connectedness. User preferences were almost evenly distributed across the three formats, with the slider-based and facial-expression-based representations each selected as the favorite by **6** participants, and the emoji-based representation by **5**. Qualitative feedback further suggested that future implementations could benefit from more integrated representations, such as avatars or hybrid formats combining interpretability with social richness.

From the author’s own observations during the sessions, the **HUD** refinements introduced in the present work particularly the text indicating the required balloon color in the archery mini-game proved effective in practice. Of the **17** participants, only **2** failed to identify the correct target color during gameplay, one of whom had removed their prescription glasses during the session, which may account for the reduced ability to perceive the on-screen text. Overall, these findings mitigate the integration approach adopted in the present work: by exposing the emotional state as a structured output consumable by external clients, the **Country Fair VR** system was designed to remain agnostic to the specific representation format, allowing results such as these to directly inform future iterations without requiring architectural changes to the core system.

4.4 SUMMARY

Taken together, the two evaluation waves document the iterative maturation of **Country Fair VR** from an early-stage usability prototype to a system ready for clinical validation. The first wave identified critical interaction issues principally the absence of tutorials and premature arrow-launch behavior all of which were systematically resolved before the principal evaluation. The second wave, conducted with **38 participants**, provided the core empirical evidence of the dissertation: the **DDA** personalization system was both functionally correct and perceptually salient (**82%** of participants reported noticing adaptation in action); the prototype achieved an **SUS score of 82** (“Excellent” category) and **98%** of participants endorsed the motivational and immersive quality of the environment. A complementary emotion-visualization study, in which the author participated as an observer, further demonstrated that all three representation formats integrated into the **Emotion Recognition Module** were well received, with the facial-expression-based format standing out in social-emotional connection, and preferences distributed evenly across the three methods. These results are now interpreted in terms of the broader research goals and theoretical commitments of this dissertation in Chapter 5.

Conclusion and Future Work

5.1 CONCLUSION

This dissertation presented **Country Fair VR**, a personalized Virtual Reality (VR) serious game prototype designed to support the upper limb and cognitive rehabilitation of stroke survivors a response to the well-documented gap in conventional rehabilitation identified in Chapter 1: the monotony and lack of adaptability inherent in traditional therapeutic exercises, which progressively erodes survivor motivation and engagement, the very pillars upon which recovery depends.

The system was designed around a conceptual framework that articulates a closed-loop architecture for personalized **VR** rehabilitation (Figure 3.4). This framework contains four interconnected components: a **VR Environment** (the Country Fair serious game scenario), a **Data Hub** responsible for comparing current performance against historical session data and implementing fatigue-check logic, a **Personalization Engine** that applies rule-based **Dynamic Difficulty Adjustment (DDA)** to maintain the survivor within the optimal state of Flow, and a **Monitoring Tool** designed for clinical oversight by healthcare professionals. What was fully implemented is the Personalization Engine, embedded in the prototype as a hybrid model that combines autonomous **DDA** with the ability for a therapist to manually override difficulty settings via a peripheral keyboard.

Within this framework, the prototype was developed around a vibrant “**Country Fair**” virtual environment. Two therapeutically grounded serious mini-games were implemented: an **Archery** mini-game, targeting bimanual static upper limb strength, proprioceptive training, and selective attention through a dual-task color matching paradigm and a **Frisbee** mini-game, targeting dynamic elbow extension, wrist coordination, visuospatial integration, and reaction capability in direct response to the flexor synergy patterns prevalent in post-stroke motor impairment. Additionally, an **Emotion Recognition Module** was integrated as a complementary component not a primary objective of this dissertation, but a proof-of-concept integration of a colleague’s parallel research. In the updated implementation, a stroke survivor’s emotional state is captured, categorized as positive, neutral, or negative, and relayed

to another survivor’s **VR** environment either as an emoji or as a valence slider contributing to the mitigation of social isolation, a critical concern identified in the clinical literature.

The prototype was evaluated through a structured, iterative user study conducted in two waves. The first wave served as an informal usability evaluation during early development, while the second and principal wave constituted a formal usability study with 38 participants spanning a wide age range and mixed levels of prior **VR** experience. The results were consistently positive across all evaluation objectives. The personalization system proved both functionally correct and perceptually salient, with the vast majority of participants recognizing the adaptive difficulty mechanism. The prototype achieved a mean **System Usability Scale (SUS)** score of **82/100** (*“Excellent”* category), and **98%** of participants described the experience as fun, motivating, and engaging. Notably, observer notes revealed that **68%** of participants spontaneously expressed a desire to continue playing after the session, providing ecological evidence of intrinsic motivation. The non-mandatory sections on acceptance and perceived therapeutic value returned exclusively positive or neutral responses, with no disagreements recorded.

5.2 FUTURE WORK

The results and limitations identified throughout this dissertation outline a clear agenda for future research and development. The following directions are proposed as the most impactful next steps for advancing **Country Fair VR** into a clinically validated rehabilitation tool:

- **Clinical validation with stroke survivors.** Conducting user studies with post-stroke survivors **Rovisco Pais Rehabilitation Center** under a formal clinical protocol is the most critical next step, as it is indispensable for translating the demonstrated motivational and usability properties of the prototype into evidence of genuine therapeutic efficacy.
- **Configurable session protocol.** Currently, the survivor selects which mini-game tent to visit freely within the fair environment, with no prescribed sequence or repetition target. Future work should allow healthcare professionals to define, prior to a session, an ordered sequence of booths and the number of repetitions required to complete each one.
- **Data Hub and session persistence.** The Data Hub defined in the conceptual framework responsible for comparing current performance against historical session data and implementing fatigue-check logic was not implemented in this dissertation. Future work should develop this processing core to enable cross-session adaptation, where the **DDA** algorithm is informed not only by in-session performance but also by the survivor’s longitudinal recovery trajectory.
- **Healthcare professional monitoring dashboard.** The Monitoring Tool outlined in the conceptual framework designed to give clinicians a structured view of each survivor’s evolution across sessions likewise falls outside the scope of this prototype and is deferred to future work. This tool should provide longitudinal visualizations of

score trajectories, difficulty level history, task completion times, and emotion state logs, enabling data-informed adjustments to the rehabilitation protocol in alignment with the hybrid oversight model preferred by healthcare professionals.

- **Expansion of the mini-game library.** Participants suggested the addition of new mini-games coherent with the Country Fair setting, such as a fishing game which would target shoulder abduction, elbow extension, and sustained grip and a duck-picking game. New mini-games should follow the same therapeutic analysis pipeline applied to the existing ones to ensure clinical relevance.

In summary, **Country Fair VR** has established a solid technical and experiential foundation for a personalized **VR** rehabilitation system. The work ahead represents a natural and well-defined roadmap toward a system that could meaningfully complement conventional stroke rehabilitation programs and contribute to the growing body of evidence supporting the therapeutic value of personalized **Extended Reality (XR)** serious games.

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Additional Content

A.1 WEBSITES OF INSTITUTIONS

- **Virtual and Augmented Reality Research Group (VAR Lab):** <https://sites.google.com/view/varlab/home>
- **Institute of Electronics and Informatics Engineering of Aveiro (IEETA):** <https://www.ieeta.pt/>
- **Rovisco Pais Rehabilitation Center:** <https://www.cm-cantanhede.pt/mcsite/Entidade/2871/Hospital-Rovisco-Pais>

A.2 ONLINE RESOURCES

- **PolyPizza:** <https://poly.pizza/>

A.3 ACADEMIC COURSES

- **Virtual And Augmented Reality Course:** <https://www.ua.pt/en/uc/12023>

A.4 COUNTRY FAIR VR PROTOTYPE DEMO VIDEOS

- **First Wave of User Tests Version:** <https://youtu.be/Yz8dQVBYweA>
- **Second Wave of User Tests Version:** <https://youtu.be/MWP1WmnOW0U>
- **Final Version:** <https://www.youtube.com/watch?v=bDXMD00PzRI>

A.5 PROJECT REPOSITORY

<https://github.com/DavidPalricas/CountryFair-VR>

A.6 SECOND WAVE OF USER TESTS QUESTIONNAIRE

CountryFair-VR-Wave-2

Section 1

Questionnaire: Evaluation of Country Fair VR

Purpose of the study

This questionnaire aims to collect information to evaluate and improve the personalization of Extended Reality (XR) experiences used in stroke rehabilitation. XR has shown strong potential to support rehabilitation by providing interactive and motivating therapeutic environments. This study seeks to better understand how these experiences can be adapted to meet the individual needs of each stroke survivor. Your participation is essential to help develop more effective, personalized, and user-centered rehabilitation solutions. All responses are anonymous and will be analysed only for academic purposes, in accordance with GDPR regulations. Thank you for your participation :)

Section 2

Demographic Data

All information provided is entirely confidential and will not be distributed or used for any purpose other than this research. Thank you!

1

Age, *

Enter your answer

2

Gender, *

☐ Male

☐ Female

☐ Other

☐ Prefer not to say

3

Professional Occupation. *

- ☐ Student
- ☐ Unemployed
- ☐ Other

4

Do you have experience with virtual reality? *



- ☐ Yes
- ☐ No

About Personalization

The next questions ask about your experience with the personalized aspects of the Virtual reality session.

5

I noticed when the system adapted the mini games during the test session. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

6

The system adapted the mini-games to my specific abilities and needs. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

Engagement & Motivation

7

The country fair environment created a fun, motivating, and engaging experience. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

8

The personalized elements increased my engagement during the session. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

9

The personalized pace helped me stay motivated throughout the activity. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

Section 5

...

Comfort & Trust

10

I did not feel overly tired or overwhelmed during the session. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

11

I trusted the system to adjust the mini-game appropriately. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

Feedback & Understanding

12

The instructions and objectives of the mini-games were clear and easy to understand. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

13

I felt that I was in a country fair. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

14

The spatial sound feedback was helpful in guiding my actions. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

Section 7

...

Acceptance

15

I believe that personalized VR could be helpful for stroke rehabilitation. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

16

I believe this experience could serve as a beneficial complement to traditional rehabilitation programs by reducing monotony. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

Effectiveness & Perceived Value

17

I believe that personalized XR could support my rehabilitation progress. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

18

The personalized experience felt beneficial for my rehabilitation. *

Strongly disagree Disagree Somewhat disagree Neutral Somewhat agree Agree Strongly agree

☐ ☐ ☐ ☐ ☐ ☐ ☐

System Usability Evaluation

19

SUS *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
I found the system unnecessarily complex.	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to be able to use .	☐	☐	☐	☐	☐
I found the various functions in the system were well integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
I imagine that most people would learn to use this product very quickly.	☐	☐	☐	☐	☐
I found the product very awkward to use.	☐	☐	☐	☐	☐
I felt very confident using the product	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this product.	☐	☐	☐	☐	☐

Suggestions or feedback

20

What did you like most and less about the game.

Enter your answer

21

Do you have any suggestions, comments, or thoughts about your experience with the personalized XR session?

Enter your answer

22

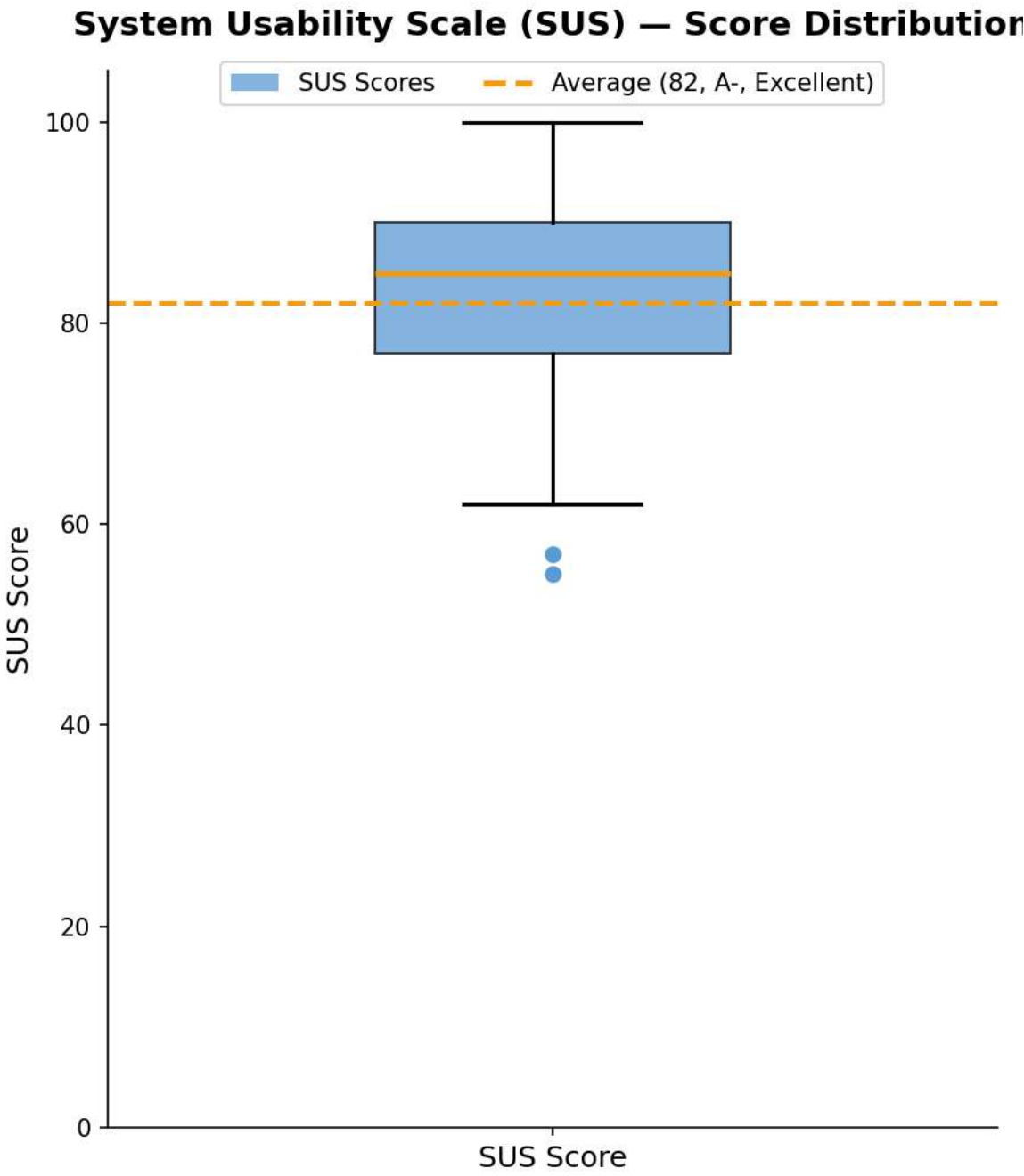
Other Suggestions

Enter your answer

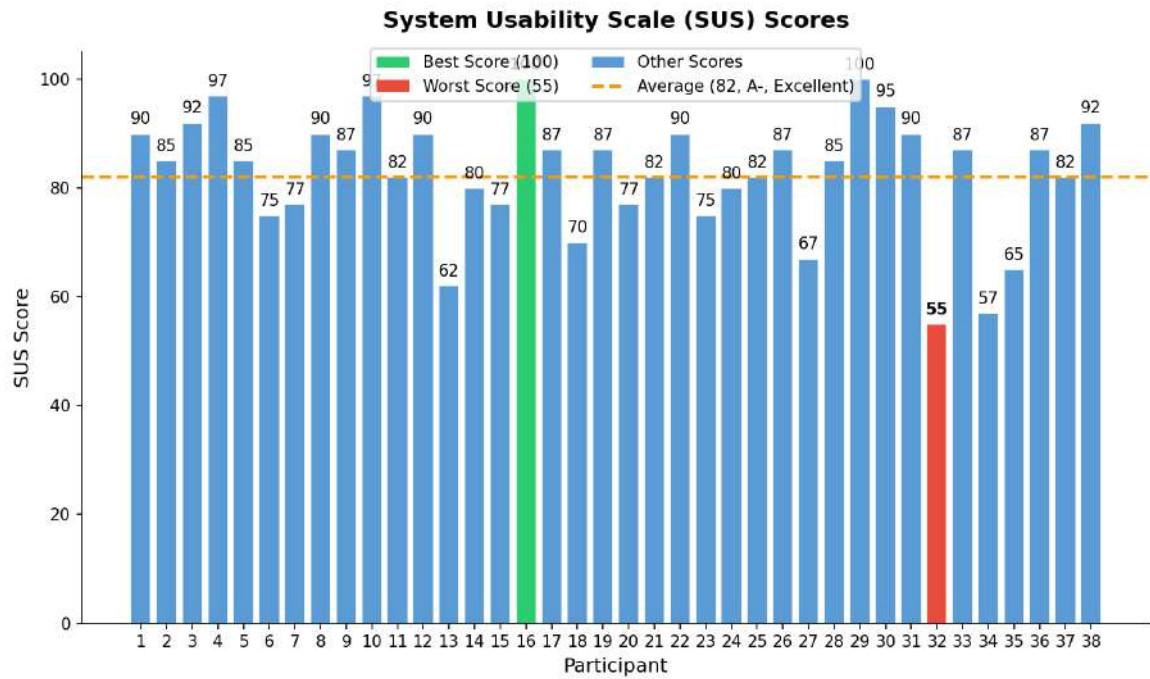
Thank you for your participation

Thank you for participating in this study! :) With your contribution, we will be able to develop Extended reality solutions that are better suited to each person's individual needs.

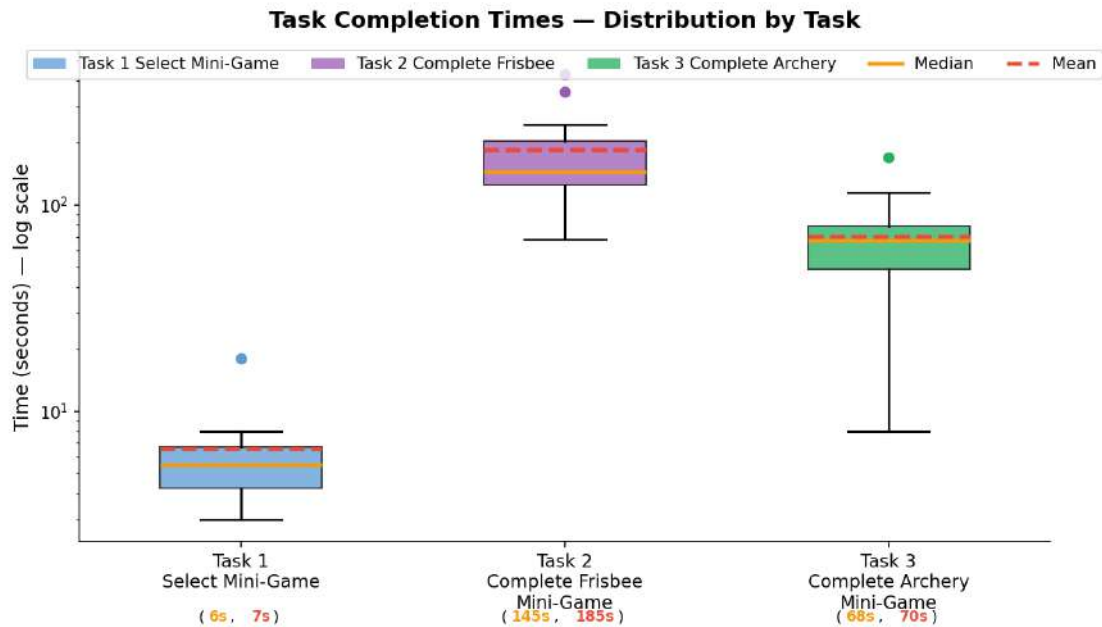
 Add new question 



A.8 SECOND WAVE OF USER TESTS SUS BARCHART



A.9 SECOND WAVE OF USER TESTS TASKS COMPLETION BOXPLOT



A.10 SECOND WAVE OF USER TESTS TASKS COMPLETION BARCHARTS

